iOS DEVELOPER INTERVIEW PRACTICE

320 Multiple-Choice Questions with Answers & Explanations

For approximately 2 years of iOS development experience

16 topic-focused sections | 4 options per question | Correct answer beneath every question

Study tip: cover each answer line while attempting the four choices. Use the answer key at the end to check performance quickly.

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Each question has one best answer. Explanations intentionally stay concise for quick revision. API details may differ by platform and deployment version.

01 Swift Fundamentals

Q1. Which description best matches let?

A. Conditionally unwraps an optional into a scoped non-optional binding.
B. Iterates over a sequence of elements.
C. Declares a binding that cannot be reassigned after initialization.
D. Inserts expressions into string text using backslash-parenthesis syntax.

Answer: C. Declares a binding that cannot be reassigned after initialization.

Q2. In iOS development, what does var mean?

A. Declares a binding whose value may be reassigned.
B. Adds a Boolean condition to a pattern or generic constraint.
C. Groups values of potentially different types without defining a new type.
D. Unwraps an optional and requires an exit from the current scope on failure.

Answer: A. Declares a binding whose value may be reassigned.

Q3. Choose the accurate description of Optional.

A. Stores ordered elements of one element type, including duplicates.
B. Represents a value that may be present or nil.
C. Matches a value against exhaustive case patterns.
D. Supplies a fallback when an optional evaluates to nil.

Answer: B. Represents a value that may be present or nil.

Q4. What is the primary purpose of if let?

A. Explicitly proceeds to the next switch case body without retesting its pattern.
B. Stores unique hashable elements with no guaranteed ordering.
C. Marks an optional type or participates in optional chaining.
D. Conditionally unwraps an optional into a scoped non-optional binding.

Answer: D. Conditionally unwraps an optional into a scoped non-optional binding.

Q5. Which description best matches guard let?

A. Schedules statements to execute when the current scope exits.
B. Extracts an optional value and traps if it is nil.
C. Unwraps an optional and requires an exit from the current scope on failure.
D. Maps unique hashable keys to associated values.

Answer: C. Unwraps an optional and requires an exit from the current scope on failure.

Q6. In iOS development, what does ?? mean?

A. Iterates over a sequence of elements.
B. Inserts expressions into string text using backslash-parenthesis syntax.
C. Supplies a fallback when an optional evaluates to nil.
D. Gives an existing type an alternative name.

Answer: C. Supplies a fallback when an optional evaluates to nil.

Q7. Choose the accurate description of ?.

A. Marks an optional type or participates in optional chaining.
B. Adds a Boolean condition to a pattern or generic constraint.
C. Groups values of potentially different types without defining a new type.
D. Represents a function or expression that does not return normally.

Answer: A. Marks an optional type or participates in optional chaining.

Q8. What is the primary purpose of ! (force unwrap)?

A. Declares a binding that cannot be reassigned after initialization.
B. Matches a value against exhaustive case patterns.
C. Extracts an optional value and traps if it is nil.
D. Stores ordered elements of one element type, including duplicates.

Answer: C. Extracts an optional value and traps if it is nil.

Q9. Which description best matches String interpolation?

A. Inserts expressions into string text using backslash-parenthesis syntax.
B. Declares a binding whose value may be reassigned.
C. Explicitly proceeds to the next switch case body without retesting its pattern.
D. Stores unique hashable elements with no guaranteed ordering.

Answer: A. Inserts expressions into string text using backslash-parenthesis syntax.

Q10. In iOS development, what does Tuple mean?

A. Maps unique hashable keys to associated values.
B. Schedules statements to execute when the current scope exits.
C. Represents a value that may be present or nil.
D. Groups values of potentially different types without defining a new type.

Answer: D. Groups values of potentially different types without defining a new type.

Q11. Choose the accurate description of Array.

A. Conditionally unwraps an optional into a scoped non-optional binding.
B. Stores ordered elements of one element type, including duplicates.
C. Gives an existing type an alternative name.
D. Iterates over a sequence of elements.

Answer: B. Stores ordered elements of one element type, including duplicates.

Q12. What is the primary purpose of Set?

A. Adds a Boolean condition to a pattern or generic constraint.
B. Represents a function or expression that does not return normally.
C. Stores unique hashable elements with no guaranteed ordering.
D. Unwraps an optional and requires an exit from the current scope on failure.

Answer: C. Stores unique hashable elements with no guaranteed ordering.

Q13. Which description best matches Dictionary?

A. Supplies a fallback when an optional evaluates to nil.

B. Maps unique hashable keys to associated values.

C. Matches a value against exhaustive case patterns.

D. Declares a binding that cannot be reassigned after initialization.

Answer: B. Maps unique hashable keys to associated values.

Q14. In iOS development, what does for-in mean?

A. Declares a binding whose value may be reassigned.

B. Marks an optional type or participates in optional chaining.

C. Iterates over a sequence of elements.

D. Explicitly proceeds to the next switch case body without retesting its pattern.

Answer: C. Iterates over a sequence of elements.

Q15. Choose the accurate description of where clause.

A. Adds a Boolean condition to a pattern or generic constraint.

B. Represents a value that may be present or nil.

C. Schedules statements to execute when the current scope exits.

D. Extracts an optional value and traps if it is nil.

Answer: A. Adds a Boolean condition to a pattern or generic constraint.

Q16. What is the primary purpose of switch?

A. Inserts expressions into string text using backslash-parenthesis syntax.

B. Conditionally unwraps an optional into a scoped non-optional binding.

C. Gives an existing type an alternative name.

D. Matches a value against exhaustive case patterns.

Answer: D. Matches a value against exhaustive case patterns.

Q17. Which description best matches fallthrough?

A. Groups values of potentially different types without defining a new type.

B. Represents a function or expression that does not return normally.

C. Unwraps an optional and requires an exit from the current scope on failure.

D. Explicitly proceeds to the next switch case body without retesting its pattern.

Answer: D. Explicitly proceeds to the next switch case body without retesting its pattern.

Q18. In iOS development, what does defer mean?

A. Schedules statements to execute when the current scope exits.

B. Declares a binding that cannot be reassigned after initialization.

C. Stores ordered elements of one element type, including duplicates.

D. Supplies a fallback when an optional evaluates to nil.

Answer: A. Schedules statements to execute when the current scope exits.

Q19. Choose the accurate description of typealias.

A. Stores unique hashable elements with no guaranteed ordering.

B. Gives an existing type an alternative name.

C. Marks an optional type or participates in optional chaining.

D. Declares a binding whose value may be reassigned.

Answer: B. Gives an existing type an alternative name.

Q20. What is the primary purpose of Never?

A. Extracts an optional value and traps if it is nil.

B. Maps unique hashable keys to associated values.

C. Represents a function or expression that does not return normally.

D. Represents a value that may be present or nil.

Answer: C. Represents a function or expression that does not return normally.

02 Value Types and OOP

Q21. Which description best matches struct?

A. Stores additional data on an individual enum case instance.
B. Runs immediately before a class instance is deallocated.
C. Declares a type-level member that cannot be overridden by subclasses.
D. Defines a value type with copy-on-write behavior for standard library collections.

Answer: D. Defines a value type with copy-on-write behavior for standard library collections.

Q22. In iOS development, what does class mean?

A. Provides a predefined literal value associated with an enum case.
B. Defines a reference type supporting inheritance and identity.
C. Declares an overridable type method on a class.
D. Checks whether two class references point to the exact same instance.

Answer: B. Defines a reference type supporting inheritance and identity.

Q23. Choose the accurate description of enum.

A. Defines a type with a finite set of named cases, possibly carrying values.
B. Enables value equality checks with the == operator.
C. Defines requirements that conforming types agree to implement.
D. Prevents overriding a member or subclassing a class.

Answer: A. Defines a type with a finite set of named cases, possibly carrying values.

Q24. What is the primary purpose of associated value?

A. Adds functionality to an existing type without subclassing it.
B. Defines a secondary class initializer that ultimately delegates to a designated initializer.
C. Stores additional data on an individual enum case instance.
D. Supports use as a Set element or Dictionary key.

Answer: C. Stores additional data on an individual enum case instance.

Q25. Which description best matches raw value?

A. Provides a predefined literal value associated with an enum case.
B. Combines Encodable and Decodable for serialization and deserialization.
C. Requires subclasses to provide or inherit a specified initializer.
D. Allows a value-type instance method to modify self or its stored properties.

Answer: A. Provides a predefined literal value associated with an enum case.

Q26. In iOS development, what does protocol mean?

A. Declares a type-level member that cannot be overridden by subclasses.
B. Provides a custom textual description property.
C. Runs immediately before a class instance is deallocated.
D. Defines requirements that conforming types agree to implement.

Answer: D. Defines requirements that conforming types agree to implement.

Q27. Choose the accurate description of extension.

A. Declares a placeholder type within a protocol definition.

B. Checks whether two class references point to the exact same instance.

C. Adds functionality to an existing type without subclassing it.

D. Declares an overridable type method on a class.

Answer: C. Adds functionality to an existing type without subclassing it.

Q28. What is the primary purpose of mutating?

A. Defines a value type with copy-on-write behavior for standard library collections.

B. Enables value equality checks with the == operator.

C. Prevents overriding a member or subclassing a class.

D. Allows a value-type instance method to modify self or its stored properties.

Answer: D. Allows a value-type instance method to modify self or its stored properties.

Q29. Which description best matches static?

A. Defines a secondary class initializer that ultimately delegates to a designated initializer.

B. Declares a type-level member that cannot be overridden by subclasses.

C. Defines a reference type supporting inheritance and identity.

D. Supports use as a Set element or Dictionary key.

Answer: B. Declares a type-level member that cannot be overridden by subclasses.

Q30. In iOS development, what does class func mean?

A. Requires subclasses to provide or inherit a specified initializer.

B. Combines Encodable and Decodable for serialization and deserialization.

C. Defines a type with a finite set of named cases, possibly carrying values.

D. Declares an overridable type method on a class.

Answer: D. Declares an overridable type method on a class.

Q31. Choose the accurate description of final.

A. Runs immediately before a class instance is deallocated.

B. Prevents overriding a member or subclassing a class.

C. Stores additional data on an individual enum case instance.

D. Provides a custom textual description property.

Answer: B. Prevents overriding a member or subclassing a class.

Q32. What is the primary purpose of convenience init?

A. Declares a placeholder type within a protocol definition.

B. Defines a secondary class initializer that ultimately delegates to a designated initializer.

C. Provides a predefined literal value associated with an enum case.

D. Checks whether two class references point to the exact same instance.

Answer: B. Defines a secondary class initializer that ultimately delegates to a designated initializer.

Q33. Which description best matches required init?

A. Requires subclasses to provide or inherit a specified initializer.
B. Enables value equality checks with the == operator.
C. Defines a value type with copy-on-write behavior for standard library collections.
D. Defines requirements that conforming types agree to implement.

Answer: A. Requires subclasses to provide or inherit a specified initializer.

Q34. In iOS development, what does deinit mean?

A. Supports use as a Set element or Dictionary key.
B. Defines a reference type supporting inheritance and identity.
C. Adds functionality to an existing type without subclassing it.
D. Runs immediately before a class instance is deallocated.

Answer: D. Runs immediately before a class instance is deallocated.

Q35. Choose the accurate description of identity operator ===.

A. Defines a type with a finite set of named cases, possibly carrying values.
B. Allows a value-type instance method to modify self or its stored properties.
C. Checks whether two class references point to the exact same instance.
D. Combines Encodable and Decodable for serialization and deserialization.

Answer: C. Checks whether two class references point to the exact same instance.

Q36. What is the primary purpose of Equatable?

A. Provides a custom textual description property.
B. Declares a type-level member that cannot be overridden by subclasses.
C. Stores additional data on an individual enum case instance.
D. Enables value equality checks with the == operator.

Answer: D. Enables value equality checks with the == operator.

Q37. Which description best matches Hashable?

A. Provides a predefined literal value associated with an enum case.
B. Declares a placeholder type within a protocol definition.
C. Supports use as a Set element or Dictionary key.
D. Declares an overridable type method on a class.

Answer: C. Supports use as a Set element or Dictionary key.

Q38. In iOS development, what does Codable mean?

A. Defines a value type with copy-on-write behavior for standard library collections.
B. Prevents overriding a member or subclassing a class.
C. Combines Encodable and Decodable for serialization and deserialization.
D. Defines requirements that conforming types agree to implement.

Answer: C. Combines Encodable and Decodable for serialization and deserialization.

Q39. Choose the accurate description of CustomStringConvertible.

A. Defines a secondary class initializer that ultimately delegates to a designated initializer.

B. Defines a reference type supporting inheritance and identity.

C. Adds functionality to an existing type without subclassing it.

D. Provides a custom textual description property.

Answer: D. Provides a custom textual description property.

Q40. What is the primary purpose of associatedtype?

A. Requires subclasses to provide or inherit a specified initializer.

B. Defines a type with a finite set of named cases, possibly carrying values.

C. Declares a placeholder type within a protocol definition.

D. Allows a value-type instance method to modify self or its stored properties.

Answer: C. Declares a placeholder type within a protocol definition.

03 Closures, Generics and Error Handling

Q41. Which description best matches closure?

A. Automatically wraps an expression in a delayed-evaluation closure.
B. A self-contained block of executable code that can capture surrounding values.
C. Maps elements to sequences then concatenates the resulting sequences.
D. Declares that a function can propagate errors.

Answer: B. A self-contained block of executable code that can capture surrounding values.

Q42. In iOS development, what does trailing closure mean?

A. Places a closure argument after a function call's parentheses.
B. Returns elements satisfying a supplied Boolean predicate.
C. Captures self weakly to avoid a strong closure-reference cycle.
D. Converts a throwing expression into an optional, returning nil when an error occurs.

Answer: A. Places a closure argument after a function call's parentheses.

Q43. Choose the accurate description of @escaping.

A. Marks a closure parameter that may be stored or invoked after the function returns.
B. Forces a throwing expression and traps if it throws.
C. Combines sequence elements into one accumulated result.
D. Captures self without retaining it and assumes it exists when used.

Answer: A. Marks a closure parameter that may be stored or invoked after the function returns.

Q44. What is the primary purpose of @autoclosure?

A. Automatically wraps an expression in a delayed-evaluation closure.
B. Uses a placeholder type parameter to work with multiple concrete types.
C. Handles thrown errors with one or more catch clauses.
D. Transforms each sequence element and returns a collection of results.

Answer: A. Automatically wraps an expression in a delayed-evaluation closure.

Q45. Which description best matches [in weak self]?

A. Captures self weakly to avoid a strong closure-reference cycle.
B. Transforms sequence elements and removes nil results.
C. Stores either a success payload or a failure payload.
D. Restricts a generic parameter to types satisfying a conformance requirement.

Answer: A. Captures self weakly to avoid a strong closure-reference cycle.

Q46. In iOS development, what does [in unowned self] mean?

A. Captures self without retaining it and assumes it exists when used.
B. Maps elements to sequences then concatenates the resulting sequences.
C. Allows a function to throw only when one of its function arguments throws.
D. Declares that a function can propagate errors.

Answer: A. Captures self without retaining it and assumes it exists when used.

Q47. Choose the accurate description of map.

A. Permits callers to ignore a function return value without a compiler warning.
B. Converts a throwing expression into an optional, returning nil when an error occurs.
C. Returns elements satisfying a supplied Boolean predicate.
D. Transforms each sequence element and returns a collection of results.

Answer: D. Transforms each sequence element and returns a collection of results.

Q48. What is the primary purpose of compactMap?

A. Transforms sequence elements and removes nil results.
B. Combines sequence elements into one accumulated result.
C. Forces a throwing expression and traps if it throws.
D. A self-contained block of executable code that can capture surrounding values.

Answer: A. Transforms sequence elements and removes nil results.

Q49. Which description best matches flatMap on sequences?

A. Uses a placeholder type parameter to work with multiple concrete types.
B. Places a closure argument after a function call's parentheses.
C. Maps elements to sequences then concatenates the resulting sequences.
D. Handles thrown errors with one or more catch clauses.

Answer: C. Maps elements to sequences then concatenates the resulting sequences.

Q50. In iOS development, what does filter mean?

A. Returns elements satisfying a supplied Boolean predicate.
B. Restricts a generic parameter to types satisfying a conformance requirement.
C. Marks a closure parameter that may be stored or invoked after the function returns.
D. Stores either a success payload or a failure payload.

Answer: A. Returns elements satisfying a supplied Boolean predicate.

Q51. Choose the accurate description of reduce.

A. Combines sequence elements into one accumulated result.
B. Declares that a function can propagate errors.
C. Automatically wraps an expression in a delayed-evaluation closure.
D. Allows a function to throw only when one of its function arguments throws.

Answer: A. Combines sequence elements into one accumulated result.

Q52. What is the primary purpose of generic function?

A. Uses a placeholder type parameter to work with multiple concrete types.
B. Captures self weakly to avoid a strong closure-reference cycle.
C. Permits callers to ignore a function return value without a compiler warning.
D. Converts a throwing expression into an optional, returning nil when an error occurs.

Answer: A. Uses a placeholder type parameter to work with multiple concrete types.

Q53. Which description best matches where T: Protocol?

A. Restricts a generic parameter to types satisfying a conformance requirement.

B. A self-contained block of executable code that can capture surrounding values.

C. Forces a throwing expression and traps if it throws.

D. Captures self without retaining it and assumes it exists when used.

Answer: A. Restricts a generic parameter to types satisfying a conformance requirement.

Q54. In iOS development, what does throws mean?

A. Handles thrown errors with one or more catch clauses.

B. Transforms each sequence element and returns a collection of results.

C. Declares that a function can propagate errors.

D. Places a closure argument after a function call's parentheses.

Answer: C. Declares that a function can propagate errors.

Q55. Choose the accurate description of try?.

A. Stores either a success payload or a failure payload.

B. Transforms sequence elements and removes nil results.

C. Marks a closure parameter that may be stored or invoked after the function returns.

D. Converts a throwing expression into an optional, returning nil when an error occurs.

Answer: D. Converts a throwing expression into an optional, returning nil when an error occurs.

Q56. What is the primary purpose of try!?

A. Maps elements to sequences then concatenates the resulting sequences.

B. Forces a throwing expression and traps if it throws.

C. Automatically wraps an expression in a delayed-evaluation closure.

D. Allows a function to throw only when one of its function arguments throws.

Answer: B. Forces a throwing expression and traps if it throws.

Q57. Which description best matches do-catch?

A. Handles thrown errors with one or more catch clauses.

B. Captures self weakly to avoid a strong closure-reference cycle.

C. Permits callers to ignore a function return value without a compiler warning.

D. Returns elements satisfying a supplied Boolean predicate.

Answer: A. Handles thrown errors with one or more catch clauses.

Q58. In iOS development, what does Result<Success, Failure> mean?

A. Combines sequence elements into one accumulated result.

B. A self-contained block of executable code that can capture surrounding values.

C. Stores either a success payload or a failure payload.

D. Captures self without retaining it and assumes it exists when used.

Answer: C. Stores either a success payload or a failure payload.

Q59. Choose the accurate description of rethrows.

A. Uses a placeholder type parameter to work with multiple concrete types.

B. Allows a function to throw only when one of its function arguments throws.

C. Places a closure argument after a function call's parentheses.

D. Transforms each sequence element and returns a collection of results.

Answer: B. Allows a function to throw only when one of its function arguments throws.

Q60. What is the primary purpose of @discardableResult?

A. Permits callers to ignore a function return value without a compiler warning.

B. Marks a closure parameter that may be stored or invoked after the function returns.

C. Restricts a generic parameter to types satisfying a conformance requirement.

D. Transforms sequence elements and removes nil results.

Answer: A. Permits callers to ignore a function return value without a compiler warning.

04 Memory Management and ARC

Q61. Which description best matches ARC?

A. Does not retain an instance and is not automatically nilled on deallocation.
B. Delays copying shared value storage until one copy is mutated.
C. Appropriate only when the closure cannot outlive the referenced instance.
D. Automatically manages class-instance lifetimes by tracking strong references.

Answer: D. Automatically manages class-instance lifetimes by tracking strong references.

Q62. In iOS development, what does strong reference mean?

A. Keeps a class instance alive while the reference exists.
B. Helps verify that an instance is released as expected.
C. Bounds the lifetime of Objective-C autoreleased objects within a scope.
D. Occurs when objects keep one another alive through strong references.

Answer: A. Keeps a class instance alive while the reference exists.

Q63. Choose the accurate description of weak reference.

A. Profiles an application for certain leaked allocations.
B. Does not retain an instance and becomes nil when its object is deallocated.
C. Common delegation ownership approach preventing the owner-delegate reference cycle.
D. Initializes stored data on first access rather than at instance creation.

Answer: B. Does not retain an instance and becomes nil when its object is deallocated.

Q64. What is the primary purpose of unowned reference?

A. Tracks heap allocations and memory usage patterns.
B. Preserves a captured value from the closure's capture point subject to capture semantics.
C. Does not retain an instance and is not automatically nilled on deallocation.
D. Specifies how values are captured by a closure.

Answer: C. Does not retain an instance and is not automatically nilled on deallocation.

Q65. Which description best matches retain cycle?

A. Safely obtains strong access to self for closure work when it remains alive.
B. Visualizes object reference chains and suspected retain cycles.
C. Occurs when objects keep one another alive through strong references.
D. May require explicit removal for block-based observation to prevent unintended retention.

Answer: C. Occurs when objects keep one another alive through strong references.

Q66. In iOS development, what does weak delegate mean?

A. Delays copying shared value storage until one copy is mutated.
B. A scheduled repeating timer can retain its target until invalidated.
C. Appropriate only when the closure cannot outlive the referenced instance.
D. Common delegation ownership approach preventing the owner-delegate reference cycle.

Answer: D. Common delegation ownership approach preventing the owner-delegate reference cycle.

Q67. Choose the accurate description of closure capture list.

A. Specifies how values are captured by a closure.

B. Helps verify that an instance is released as expected.

C. Bounds the lifetime of Objective-C autoreleased objects within a scope.

D. A long-running task may prolong an object's lifetime when it strongly captures self.

Answer: A. Specifies how values are captured by a closure.

Q68. What is the primary purpose of [weak self] guard let self?

A. Initializes stored data on first access rather than at instance creation.

B. Automatically manages class-instance lifetimes by tracking strong references.

C. Safely obtains strong access to self for closure work when it remains alive.

D. Profiles an application for certain leaked allocations.

Answer: C. Safely obtains strong access to self for closure work when it remains alive.

Q69. Which description best matches [unowned self]?

A. Appropriate only when the closure cannot outlive the referenced instance.

B. Tracks heap allocations and memory usage patterns.

C. Keeps a class instance alive while the reference exists.

D. Preserves a captured value from the closure's capture point subject to capture semantics.

Answer: A. Appropriate only when the closure cannot outlive the referenced instance.

Q70. In iOS development, what does deinit breakpoint mean?

A. Does not retain an instance and becomes nil when its object is deallocated.

B. Visualizes object reference chains and suspected retain cycles.

C. May require explicit removal for block-based observation to prevent unintended retention.

D. Helps verify that an instance is released as expected.

Answer: D. Helps verify that an instance is released as expected.

Q71. Choose the accurate description of Instruments Leaks.

A. Delays copying shared value storage until one copy is mutated.

B. Profiles an application for certain leaked allocations.

C. A scheduled repeating timer can retain its target until invalidated.

D. Does not retain an instance and is not automatically nilled on deallocation.

Answer: B. Profiles an application for certain leaked allocations.

Q72. What is the primary purpose of Instruments Allocations?

A. A long-running task may prolong an object's lifetime when it strongly captures self.

B. Occurs when objects keep one another alive through strong references.

C. Bounds the lifetime of Objective-C autoreleased objects within a scope.

D. Tracks heap allocations and memory usage patterns.

Answer: D. Tracks heap allocations and memory usage patterns.

Q73. Which description best matches Xcode Memory Graph?

A. Initializes stored data on first access rather than at instance creation.

B. Common delegation ownership approach preventing the owner-delegate reference cycle.

C. Visualizes object reference chains and suspected retain cycles.

D. Automatically manages class-instance lifetimes by tracking strong references.

Answer: C. Visualizes object reference chains and suspected retain cycles.

Q74. In iOS development, what does copy-on-write mean?

A. Preserves a captured value from the closure's capture point subject to capture semantics.

B. Keeps a class instance alive while the reference exists.

C. Delays copying shared value storage until one copy is mutated.

D. Specifies how values are captured by a closure.

Answer: C. Delays copying shared value storage until one copy is mutated.

Q75. Choose the accurate description of autoreleasepool.

A. Safely obtains strong access to self for closure work when it remains alive.

B. May require explicit removal for block-based observation to prevent unintended retention.

C. Does not retain an instance and becomes nil when its object is deallocated.

D. Bounds the lifetime of Objective-C autoreleased objects within a scope.

Answer: D. Bounds the lifetime of Objective-C autoreleased objects within a scope.

Q76. What is the primary purpose of lazy stored property?

A. A scheduled repeating timer can retain its target until invalidated.

B. Appropriate only when the closure cannot outlive the referenced instance.

C. Does not retain an instance and is not automatically nilled on deallocation.

D. Initializes stored data on first access rather than at instance creation.

Answer: D. Initializes stored data on first access rather than at instance creation.

Q77. Which description best matches capture by value?

A. Preserves a captured value from the closure's capture point subject to capture semantics.

B. A long-running task may prolong an object's lifetime when it strongly captures self.

C. Occurs when objects keep one another alive through strong references.

D. Helps verify that an instance is released as expected.

Answer: A. Preserves a captured value from the closure's capture point subject to capture semantics.

Q78. In iOS development, what does NotificationCenter observer token mean?

A. Common delegation ownership approach preventing the owner-delegate reference cycle.

B. Automatically manages class-instance lifetimes by tracking strong references.

C. Profiles an application for certain leaked allocations.

D. May require explicit removal for block-based observation to prevent unintended retention.

Answer: D. May require explicit removal for block-based observation to prevent unintended retention.

Q79. Choose the accurate description of Timer target ownership.

A. A scheduled repeating timer can retain its target until invalidated.

B. Tracks heap allocations and memory usage patterns.

C. Specifies how values are captured by a closure.

D. Keeps a class instance alive while the reference exists.

Answer: A. A scheduled repeating timer can retain its target until invalidated.

Q80. What is the primary purpose of Task closure capture?

A. Visualizes object reference chains and suspected retain cycles.

B. Safely obtains strong access to self for closure work when it remains alive.

C. A long-running task may prolong an object's lifetime when it strongly captures self.

D. Does not retain an instance and becomes nil when its object is deallocated.

Answer: C. A long-running task may prolong an object's lifetime when it strongly captures self.

05 UIKit Fundamentals

Q81. Which description best matches UIView?

A. Represents a rectangular region responsible for drawing and event handling.
B. Called shortly before the view becomes visible.
C. Manages a stack of view controllers for hierarchical navigation.
D. Displays data-driven items in a customizable layout.

Answer: A. Represents a rectangular region responsible for drawing and event handling.

Q82. In iOS development, what does UIViewController mean?

A. Called after the view has appeared onscreen.
B. Manages a screen's view hierarchy and lifecycle.
C. Manages multiple selectable child view controllers via tabs.
D. Displays vertically scrolling rows organized into sections.

Answer: B. Manages a screen's view hierarchy and lifecycle.

Q83. Choose the accurate description of viewDidLoad.

A. Called before the view is removed from visibility.
B. Displays a view controller modally.
C. Recycles offscreen cell instances to reduce allocation overhead.
D. Called after the controller's view is loaded into memory.

Answer: D. Called after the controller's view is loaded into memory.

Q84. What is the primary purpose of viewWillAppear?

A. Identifies a typical button tap event on release inside its bounds.
B. Adds a controller onto a navigation controller stack.
C. Called shortly before the view becomes visible.
D. Called after the view is no longer visible.

Answer: C. Called shortly before the view becomes visible.

Q85. Which description best matches viewDidAppear?

A. Connects a storyboard or nib interface object to code.
B. Called after the view has appeared onscreen.
C. Arranges child views along a horizontal or vertical axis.
D. Creates the controller's view programmatically when not using a storyboard or nib.

Answer: B. Called after the view has appeared onscreen.

Q86. In iOS development, what does viewWillDisappear mean?

A. Displays data-driven items in a customizable layout.
B. Manages a stack of view controllers for hierarchical navigation.
C. Called before the view is removed from visibility.
D. Connects a storyboard or nib user action to a method.

Answer: C. Called before the view is removed from visibility.

Q87. Choose the accurate description of viewDidDisappear.

A. Displays vertically scrolling rows organized into sections.

B. Routes unhandled events or actions through responder objects.

C. Called after the view is no longer visible.

D. Manages multiple selectable child view controllers via tabs.

Answer: C. Called after the view is no longer visible.

Q88. What is the primary purpose of loadView?

A. Represents a rectangular region responsible for drawing and event handling.

B. Creates the controller's view programmatically when not using a storyboard or nib.

C. Recycles offscreen cell instances to reduce allocation overhead.

D. Displays a view controller modally.

Answer: B. Creates the controller's view programmatically when not using a storyboard or nib.

Q89. Which description best matches UINavigationController?

A. Identifies a typical button tap event on release inside its bounds.

B. Manages a stack of view controllers for hierarchical navigation.

C. Manages a screen's view hierarchy and lifecycle.

D. Adds a controller onto a navigation controller stack.

Answer: B. Manages a stack of view controllers for hierarchical navigation.

Q90. In iOS development, what does UITabBarController mean?

A. Connects a storyboard or nib interface object to code.

B. Manages multiple selectable child view controllers via tabs.

C. Arranges child views along a horizontal or vertical axis.

D. Called after the controller's view is loaded into memory.

Answer: B. Manages multiple selectable child view controllers via tabs.

Q91. Choose the accurate description of present(_:animated:).

A. Connects a storyboard or nib user action to a method.

B. Displays data-driven items in a customizable layout.

C. Called shortly before the view becomes visible.

D. Displays a view controller modally.

Answer: D. Displays a view controller modally.

Q92. What is the primary purpose of pushViewController?

A. Called after the view has appeared onscreen.

B. Displays vertically scrolling rows organized into sections.

C. Adds a controller onto a navigation controller stack.

D. Routes unhandled events or actions through responder objects.

Answer: C. Adds a controller onto a navigation controller stack.

Q93. Which description best matches UIStackView?

A. Represents a rectangular region responsible for drawing and event handling.

B. Arranges child views along a horizontal or vertical axis.

C. Called before the view is removed from visibility.

D. Recycles offscreen cell instances to reduce allocation overhead.

Answer: B. Arranges child views along a horizontal or vertical axis.

Q94. In iOS development, what does UICollectionView mean?

A. Identifies a typical button tap event on release inside its bounds.

B. Manages a screen's view hierarchy and lifecycle.

C. Called after the view is no longer visible.

D. Displays data-driven items in a customizable layout.

Answer: D. Displays data-driven items in a customizable layout.

Q95. Choose the accurate description of UITableView.

A. Connects a storyboard or nib interface object to code.

B. Displays vertically scrolling rows organized into sections.

C. Creates the controller's view programmatically when not using a storyboard or nib.

D. Called after the controller's view is loaded into memory.

Answer: B. Displays vertically scrolling rows organized into sections.

Q96. What is the primary purpose of UITableViewCell reuse?

A. Manages a stack of view controllers for hierarchical navigation.

B. Connects a storyboard or nib user action to a method.

C. Called shortly before the view becomes visible.

D. Recycles offscreen cell instances to reduce allocation overhead.

Answer: D. Recycles offscreen cell instances to reduce allocation overhead.

Q97. Which description best matches UIControl.Event.touchUpInside?

A. Routes unhandled events or actions through responder objects.

B. Called after the view has appeared onscreen.

C. Manages multiple selectable child view controllers via tabs.

D. Identifies a typical button tap event on release inside its bounds.

Answer: D. Identifies a typical button tap event on release inside its bounds.

Q98. In iOS development, what does IBOutlet mean?

A. Called before the view is removed from visibility.

B. Connects a storyboard or nib interface object to code.

C. Represents a rectangular region responsible for drawing and event handling.

D. Displays a view controller modally.

Answer: B. Connects a storyboard or nib interface object to code.

Q99. Choose the accurate description of IBAction.

A. Manages a screen's view hierarchy and lifecycle.

B. Called after the view is no longer visible.

C. Adds a controller onto a navigation controller stack.

D. Connects a storyboard or nib user action to a method.

Answer: D. Connects a storyboard or nib user action to a method.

Q100. What is the primary purpose of UIResponder chain?

A. Creates the controller's view programmatically when not using a storyboard or nib.

B. Routes unhandled events or actions through responder objects.

C. Arranges child views along a horizontal or vertical axis.

D. Called after the controller's view is loaded into memory.

Answer: B. Routes unhandled events or actions through responder objects.

06 Auto Layout and UI Performance

Q101. Which description best matches Auto Layout constraint?

A. Defines a relationship between layout attributes such as widths or positions.
B. Requests pending layout changes to be applied immediately where possible.
C. Provides the view's preferred inset layout region.
D. Requires UIKit view hierarchy changes to take place on the main thread.

Answer: A. Defines a relationship between layout attributes such as widths or positions.

Q102. In iOS development, what does translatesAutoresizingMaskIntoConstraints = false mean?

A. Marks a view as needing to be redrawn on a future display pass.
B. Controls how strongly a view resists growing beyond intrinsic content size.
C. Requires updating all reused cell state to avoid showing stale content.
D. Prevents automatic conversion of autoresizing masks into constraints.

Answer: D. Prevents automatic conversion of autoresizing masks into constraints.

Q103. Choose the accurate description of safeAreaLayoutGuide.

A. Applies snapshot-based updates to list and collection interfaces.
B. Controls how strongly a view resists shrinking below intrinsic content size.
C. Allows table views to estimate cell heights while performing layout.
D. Represents an area that avoids bars, notches and other system-obscured regions.

Answer: D. Represents an area that avoids bars, notches and other system-obscured regions.

Q104. What is the primary purpose of layoutMarginsGuide?

A. Provides a pull-to-refresh interaction for scrollable content.
B. Provides the view's preferred inset layout region.
C. Reports a view's natural size based on its content.
D. Starts preparing data before cells become visible.

Answer: B. Provides the view's preferred inset layout region.

Q105. Which description best matches content hugging priority?

A. Legacy callback for responding to changes in size-class or appearance traits.
B. Decodes images near their display size to reduce memory and decoding cost.
C. Marks a view as requiring a future layout pass.
D. Controls how strongly a view resists growing beyond intrinsic content size.

Answer: D. Controls how strongly a view resists growing beyond intrinsic content size.

Q106. In iOS development, what does compression resistance priority mean?

A. Requests pending layout changes to be applied immediately where possible.
B. Requires UIKit view hierarchy changes to take place on the main thread.
C. Controls how strongly a view resists shrinking below intrinsic content size.
D. Provides a modern way to configure cell content.

Answer: C. Controls how strongly a view resists shrinking below intrinsic content size.

Q107. Choose the accurate description of intrinsicContentSize.

A. Requires updating all reused cell state to avoid showing stale content.

B. Marks a view as needing to be redrawn on a future display pass.

C. Reports a view's natural size based on its content.

D. Applies supported appearance customization through proxy objects.

Answer: C. Reports a view's natural size based on its content.

Q108. What is the primary purpose of setNeedsLayout?

A. Marks a view as requiring a future layout pass.

B. Defines a relationship between layout attributes such as widths or positions.

C. Applies snapshot-based updates to list and collection interfaces.

D. Allows table views to estimate cell heights while performing layout.

Answer: A. Marks a view as requiring a future layout pass.

Q109. Which description best matches layoutIfNeeded?

A. Provides a pull-to-refresh interaction for scrollable content.

B. Starts preparing data before cells become visible.

C. Requests pending layout changes to be applied immediately where possible.

D. Prevents automatic conversion of autoresizing masks into constraints.

Answer: C. Requests pending layout changes to be applied immediately where possible.

Q110. In iOS development, what does setNeedsDisplay mean?

A. Decodes images near their display size to reduce memory and decoding cost.

B. Legacy callback for responding to changes in size-class or appearance traits.

C. Marks a view as needing to be redrawn on a future display pass.

D. Represents an area that avoids bars, notches and other system-obscured regions.

Answer: C. Marks a view as needing to be redrawn on a future display pass.

Q111. Choose the accurate description of Diffable data source.

A. Requires UIKit view hierarchy changes to take place on the main thread.

B. Provides a modern way to configure cell content.

C. Provides the view's preferred inset layout region.

D. Applies snapshot-based updates to list and collection interfaces.

Answer: D. Applies snapshot-based updates to list and collection interfaces.

Q112. What is the primary purpose of prefetching?

A. Requires updating all reused cell state to avoid showing stale content.

B. Controls how strongly a view resists growing beyond intrinsic content size.

C. Applies supported appearance customization through proxy objects.

D. Starts preparing data before cells become visible.

Answer: D. Starts preparing data before cells become visible.

Q113. Which description best matches image downsampling?

A. Decodes images near their display size to reduce memory and decoding cost.

B. Defines a relationship between layout attributes such as widths or positions.

C. Controls how strongly a view resists shrinking below intrinsic content size.

D. Allows table views to estimate cell heights while performing layout.

Answer: A. Decodes images near their display size to reduce memory and decoding cost.

Q114. In iOS development, what does main thread UI rule mean?

A. Requires UIKit view hierarchy changes to take place on the main thread.

B. Provides a pull-to-refresh interaction for scrollable content.

C. Prevents automatic conversion of autoresizing masks into constraints.

D. Reports a view's natural size based on its content.

Answer: A. Requires UIKit view hierarchy changes to take place on the main thread.

Q115. Choose the accurate description of cell configuration reuse.

A. Requires updating all reused cell state to avoid showing stale content.

B. Marks a view as requiring a future layout pass.

C. Represents an area that avoids bars, notches and other system-obscured regions.

D. Legacy callback for responding to changes in size-class or appearance traits.

Answer: A. Requires updating all reused cell state to avoid showing stale content.

Q116. What is the primary purpose of estimatedRowHeight?

A. Allows table views to estimate cell heights while performing layout.

B. Requests pending layout changes to be applied immediately where possible.

C. Provides a modern way to configure cell content.

D. Provides the view's preferred inset layout region.

Answer: A. Allows table views to estimate cell heights while performing layout.

Q117. Which description best matches UIRefreshControl?

A. Controls how strongly a view resists growing beyond intrinsic content size.

B. Applies supported appearance customization through proxy objects.

C. Marks a view as needing to be redrawn on a future display pass.

D. Provides a pull-to-refresh interaction for scrollable content.

Answer: D. Provides a pull-to-refresh interaction for scrollable content.

Q118. In iOS development, what does traitCollectionDidChange mean?

A. Applies snapshot-based updates to list and collection interfaces.

B. Defines a relationship between layout attributes such as widths or positions.

C. Legacy callback for responding to changes in size-class or appearance traits.

D. Controls how strongly a view resists shrinking below intrinsic content size.

Answer: C. Legacy callback for responding to changes in size-class or appearance traits.

Q119. Choose the accurate description of UIContentConfiguration.

A. Prevents automatic conversion of autoresizing masks into constraints.

B. Provides a modern way to configure cell content.

C. Starts preparing data before cells become visible.

D. Reports a view's natural size based on its content.

Answer: B. Provides a modern way to configure cell content.

Q120. What is the primary purpose of UIAppearance?

A. Decodes images near their display size to reduce memory and decoding cost.

B. Applies supported appearance customization through proxy objects.

C. Represents an area that avoids bars, notches and other system-obscured regions.

D. Marks a view as requiring a future layout pass.

Answer: B. Applies supported appearance customization through proxy objects.

07 SwiftUI Essentials

Q121. Which description best matches View protocol?

A. Creates and preserves an ObservableObject across view redraws in legacy observation code.
B. Emits change notifications for a stored property of an ObservableObject.
C. Defines a description of part of a SwiftUI interface through its body.
D. Creates child views from an identifiable or explicitly identified collection.

Answer: C. Defines a description of part of a SwiftUI interface through its body.

Q122. In iOS development, what does @State mean?

A. Owns simple mutable view-local state and triggers updates when it changes.
B. Observes an externally owned ObservableObject for changes.
C. Macro that enables modern observation of accessed model properties.
D. Builds stack children lazily as they approach visibility.

Answer: A. Owns simple mutable view-local state and triggers updates when it changes.

Q123. Choose the accurate description of @Binding.

A. Reads a shared ObservableObject injected into the view environment.
B. Exposes available layout geometry to its content closure.
C. Provides two-way access to state owned elsewhere.
D. Manages data-driven navigation through a stack of destinations.

Answer: C. Provides two-way access to state owned elsewhere.

Q124. What is the primary purpose of @StateObject?

A. Runs an action when a view appears in the interface.
B. Reads a value provided through the SwiftUI environment.
C. Creates and preserves an ObservableObject across view redraws in legacy observation code.
D. Activates navigation to a destination view or value.

Answer: C. Creates and preserves an ObservableObject across view redraws in legacy observation code.

Q125. Which description best matches @ObservedObject?

A. Runs asynchronous work tied to the view lifecycle.
B. Displays structured, scrollable rows with platform list behavior.
C. Observes an externally owned ObservableObject for changes.
D. Protocol whose change publisher can announce updates to observing views.

Answer: C. Observes an externally owned ObservableObject for changes.

Q126. In iOS development, what does @EnvironmentObject mean?

A. Presents a modal sheet based on a binding or item.
B. Emits change notifications for a stored property of an ObservableObject.
C. Reads a shared ObservableObject injected into the view environment.
D. Creates child views from an identifiable or explicitly identified collection.

Answer: C. Reads a shared ObservableObject injected into the view environment.

Q127. Choose the accurate description of @Environment.

A. Builds stack children lazily as they approach visibility.

B. Macro that enables modern observation of accessed model properties.

C. Requests a preferred light or dark appearance for the view hierarchy.

D. Reads a value provided through the SwiftUI environment.

Answer: D. Reads a value provided through the SwiftUI environment.

Q128. What is the primary purpose of ObservableObject?

A. Protocol whose change publisher can announce updates to observing views.

B. Defines a description of part of a SwiftUI interface through its body.

C. Manages data-driven navigation through a stack of destinations.

D. Exposes available layout geometry to its content closure.

Answer: A. Protocol whose change publisher can announce updates to observing views.

Q129. Which description best matches @Published?

A. Activates navigation to a destination view or value.

B. Runs an action when a view appears in the interface.

C. Owns simple mutable view-local state and triggers updates when it changes.

D. Emits change notifications for a stored property of an ObservableObject.

Answer: D. Emits change notifications for a stored property of an ObservableObject.

Q130. In iOS development, what does @Observable mean?

A. Displays structured, scrollable rows with platform list behavior.

B. Runs asynchronous work tied to the view lifecycle.

C. Provides two-way access to state owned elsewhere.

D. Macro that enables modern observation of accessed model properties.

Answer: D. Macro that enables modern observation of accessed model properties.

Q131. Choose the accurate description of NavigationStack.

A. Presents a modal sheet based on a binding or item.

B. Creates child views from an identifiable or explicitly identified collection.

C. Creates and preserves an ObservableObject across view redraws in legacy observation code.

D. Manages data-driven navigation through a stack of destinations.

Answer: D. Manages data-driven navigation through a stack of destinations.

Q132. What is the primary purpose of NavigationLink?

A. Activates navigation to a destination view or value.

B. Builds stack children lazily as they approach visibility.

C. Requests a preferred light or dark appearance for the view hierarchy.

D. Observes an externally owned ObservableObject for changes.

Answer: A. Activates navigation to a destination view or value.

Q133. Which description best matches List?

A. Displays structured, scrollable rows with platform list behavior.

B. Defines a description of part of a SwiftUI interface through its body.

C. Reads a shared ObservableObject injected into the view environment.

D. Exposes available layout geometry to its content closure.

Answer: A. Displays structured, scrollable rows with platform list behavior.

Q134. In iOS development, what does ForEach mean?

A. Owns simple mutable view-local state and triggers updates when it changes.

B. Runs an action when a view appears in the interface.

C. Reads a value provided through the SwiftUI environment.

D. Creates child views from an identifiable or explicitly identified collection.

Answer: D. Creates child views from an identifiable or explicitly identified collection.

Q135. Choose the accurate description of LazyVStack.

A. Provides two-way access to state owned elsewhere.

B. Builds stack children lazily as they approach visibility.

C. Protocol whose change publisher can announce updates to observing views.

D. Runs asynchronous work tied to the view lifecycle.

Answer: B. Builds stack children lazily as they approach visibility.

Q136. What is the primary purpose of GeometryReader?

A. Creates and preserves an ObservableObject across view redraws in legacy observation code.

B. Exposes available layout geometry to its content closure.

C. Presents a modal sheet based on a binding or item.

D. Emits change notifications for a stored property of an ObservableObject.

Answer: B. Exposes available layout geometry to its content closure.

Q137. Which description best matches .onAppear?

A. Macro that enables modern observation of accessed model properties.

B. Runs an action when a view appears in the interface.

C. Requests a preferred light or dark appearance for the view hierarchy.

D. Observes an externally owned ObservableObject for changes.

Answer: B. Runs an action when a view appears in the interface.

Q138. In iOS development, what does .task mean?

A. Defines a description of part of a SwiftUI interface through its body.

B. Manages data-driven navigation through a stack of destinations.

C. Runs asynchronous work tied to the view lifecycle.

D. Reads a shared ObservableObject injected into the view environment.

Answer: C. Runs asynchronous work tied to the view lifecycle.

Q139. Choose the accurate description of .sheet.

A. Activates navigation to a destination view or value.

B. Presents a modal sheet based on a binding or item.

C. Reads a value provided through the SwiftUI environment.

D. Owns simple mutable view-local state and triggers updates when it changes.

Answer: B. Presents a modal sheet based on a binding or item.

Q140. What is the primary purpose of .preferredColorScheme?

A. Displays structured, scrollable rows with platform list behavior.

B. Protocol whose change publisher can announce updates to observing views.

C. Provides two-way access to state owned elsewhere.

D. Requests a preferred light or dark appearance for the view hierarchy.

Answer: D. Requests a preferred light or dark appearance for the view hierarchy.

08 Concurrency and Threading

Q141. Which description best matches async?

A. Marks a function that may suspend while doing asynchronous work.
B. Creates an unstructured task without inheriting the current actor context.
C. Schedules a block asynchronously on the main dispatch queue.
D. Cooperative request indicating that asynchronous work should stop.

Answer: A. Marks a function that may suspend while doing asynchronous work.

Q142. In iOS development, what does await mean?

A. Marks a potential suspension point when invoking asynchronous code.
B. Global actor used to isolate UI-related work to the main execution context.
C. Accesses a system-managed concurrent queue for background work.
D. Throws a cancellation error if the current task was cancelled.

Answer: A. Marks a potential suspension point when invoking asynchronous code.

Q143. Choose the accurate description of Task.

A. Executes one submitted task at a time in queue order.
B. Protects mutable state through actor isolation.
C. Reports whether the current task has a cancellation request.
D. Creates a unit of asynchronous work managed by Swift concurrency.

Answer: D. Creates a unit of asynchronous work managed by Swift concurrency.

Q144. What is the primary purpose of Task.detached?

A. Runs dynamically added child tasks in structured concurrency.
B. Can execute multiple independent submitted tasks at the same time.
C. Creates an unstructured task without inheriting the current actor context.
D. Bridges one-shot callback APIs into async code with continuation checks.

Answer: C. Creates an unstructured task without inheriting the current actor context.

Q145. Which description best matches MainActor?

A. Bridges throwing callback APIs into async code.
B. Occurs when execution cannot proceed because participants wait indefinitely on one another.
C. Global actor used to isolate UI-related work to the main execution context.
D. Starts a fixed set of concurrent child operations within a scope.

Answer: C. Global actor used to isolate UI-related work to the main execution context.

Q146. In iOS development, what does actor mean?

A. Occurs when an outcome depends on uncontrolled interleaving of concurrent operations.
B. Cooperative request indicating that asynchronous work should stop.
C. Protects mutable state through actor isolation.
D. Schedules a block asynchronously on the main dispatch queue.

Answer: C. Protects mutable state through actor isolation.

Q147. Choose the accurate description of TaskGroup.

A. Runs dynamically added child tasks in structured concurrency.

B. Accesses a system-managed concurrent queue for background work.

C. Throws a cancellation error if the current task was cancelled.

D. Marks values considered safe to transfer across concurrency boundaries.

Answer: A. Runs dynamically added child tasks in structured concurrency.

Q148. What is the primary purpose of async let?

A. Marks a function that may suspend while doing asynchronous work.

B. Starts a fixed set of concurrent child operations within a scope.

C. Executes one submitted task at a time in queue order.

D. Reports whether the current task has a cancellation request.

Answer: B. Starts a fixed set of concurrent child operations within a scope.

Q149. Which description best matches Task cancellation?

A. Marks a potential suspension point when invoking asynchronous code.

B. Cooperative request indicating that asynchronous work should stop.

C. Can execute multiple independent submitted tasks at the same time.

D. Bridges one-shot callback APIs into async code with continuation checks.

Answer: B. Cooperative request indicating that asynchronous work should stop.

Q150. In iOS development, what does Task.checkCancellation() mean?

A. Occurs when execution cannot proceed because participants wait indefinitely on one another.

B. Bridges throwing callback APIs into async code.

C. Throws a cancellation error if the current task was cancelled.

D. Creates a unit of asynchronous work managed by Swift concurrency.

Answer: C. Throws a cancellation error if the current task was cancelled.

Q151. Choose the accurate description of Task.isCancelled.

A. Creates an unstructured task without inheriting the current actor context.

B. Occurs when an outcome depends on uncontrolled interleaving of concurrent operations.

C. Reports whether the current task has a cancellation request.

D. Schedules a block asynchronously on the main dispatch queue.

Answer: C. Reports whether the current task has a cancellation request.

Q152. What is the primary purpose of withCheckedContinuation?

A. Global actor used to isolate UI-related work to the main execution context.

B. Bridges one-shot callback APIs into async code with continuation checks.

C. Marks values considered safe to transfer across concurrency boundaries.

D. Accesses a system-managed concurrent queue for background work.

Answer: B. Bridges one-shot callback APIs into async code with continuation checks.

Q153. Which description best matches withCheckedThrowingContinuation?

A. Protects mutable state through actor isolation.

B. Marks a function that may suspend while doing asynchronous work.

C. Bridges throwing callback APIs into async code.

D. Executes one submitted task at a time in queue order.

Answer: C. Bridges throwing callback APIs into async code.

Q154. In iOS development, what does DispatchQueue.main.async mean?

A. Schedules a block asynchronously on the main dispatch queue.

B. Marks a potential suspension point when invoking asynchronous code.

C. Can execute multiple independent submitted tasks at the same time.

D. Runs dynamically added child tasks in structured concurrency.

Answer: A. Schedules a block asynchronously on the main dispatch queue.

Q155. Choose the accurate description of DispatchQueue.global().

A. Starts a fixed set of concurrent child operations within a scope.

B. Accesses a system-managed concurrent queue for background work.

C. Occurs when execution cannot proceed because participants wait indefinitely on one another.

D. Creates a unit of asynchronous work managed by Swift concurrency.

Answer: B. Accesses a system-managed concurrent queue for background work.

Q156. What is the primary purpose of serial queue?

A. Occurs when an outcome depends on uncontrolled interleaving of concurrent operations.

B. Executes one submitted task at a time in queue order.

C. Cooperative request indicating that asynchronous work should stop.

D. Creates an unstructured task without inheriting the current actor context.

Answer: B. Executes one submitted task at a time in queue order.

Q157. Which description best matches concurrent queue?

A. Can execute multiple independent submitted tasks at the same time.

B. Global actor used to isolate UI-related work to the main execution context.

C. Throws a cancellation error if the current task was cancelled.

D. Marks values considered safe to transfer across concurrency boundaries.

Answer: A. Can execute multiple independent submitted tasks at the same time.

Q158. In iOS development, what does deadlock mean?

A. Reports whether the current task has a cancellation request.

B. Marks a function that may suspend while doing asynchronous work.

C. Protects mutable state through actor isolation.

D. Occurs when execution cannot proceed because participants wait indefinitely on one another.

Answer: D. Occurs when execution cannot proceed because participants wait indefinitely on one another.

Q159. Choose the accurate description of race condition.

A. Occurs when an outcome depends on uncontrolled interleaving of concurrent operations.

B. Marks a potential suspension point when invoking asynchronous code.

C. Bridges one-shot callback APIs into async code with continuation checks.

D. Runs dynamically added child tasks in structured concurrency.

Answer: A. Occurs when an outcome depends on uncontrolled interleaving of concurrent operations.

Q160. What is the primary purpose of Sendable?

A. Marks values considered safe to transfer across concurrency boundaries.

B. Starts a fixed set of concurrent child operations within a scope.

C. Bridges throwing callback APIs into async code.

D. Creates a unit of asynchronous work managed by Swift concurrency.

Answer: A. Marks values considered safe to transfer across concurrency boundaries.

09 Networking and REST APIs

Q161. Which description best matches URLSession?

A. Indicates the requested resource was not found.
B. Apple's framework API for HTTP networking and data transfer.
C. Submits data for server-side processing or resource creation.
D. Indicates a resource was created successfully.

Answer: B. Apple's framework API for HTTP networking and data transfer.

Q162. In iOS development, what does URLRequest mean?

A. Typically replaces the representation of a specified resource.
B. Indicates a client has sent too many requests.
C. Indicates success without a response body.
D. Stores request URL, method, headers and optional body.

Answer: D. Stores request URL, method, headers and optional body.

Q163. Choose the accurate description of HTTP GET.

A. Indicates a generic internal server error.
B. Retrieves a resource without intending to modify server state.
C. Indicates a client-side bad request.
D. Applies partial modifications to a resource.

Answer: B. Retrieves a resource without intending to modify server state.

Q164. What is the primary purpose of HTTP POST?

A. Submits data for server-side processing or resource creation.
B. Requests removal of a resource.
C. Declares the media type of a JSON request or response body.
D. Indicates missing or invalid authentication credentials.

Answer: A. Submits data for server-side processing or resource creation.

Q165. Which description best matches HTTP PUT?

A. Indicates the server understood the request but refuses authorization.
B. Typically replaces the representation of a specified resource.
C. Carries a bearer access token in an HTTP authorization header.
D. Indicates a successful request response.

Answer: B. Typically replaces the representation of a specified resource.

Q166. In iOS development, what does HTTP PATCH mean?

A. Safely assembles URLs and percent-encoded query items.
B. Indicates the requested resource was not found.
C. Applies partial modifications to a resource.
D. Indicates a resource was created successfully.

Answer: C. Applies partial modifications to a resource.

Q167. Choose the accurate description of HTTP DELETE.

A. Indicates a client has sent too many requests.

B. Performs a URL request and returns data plus response asynchronously.

C. Indicates success without a response body.

D. Requests removal of a resource.

Answer: D. Requests removal of a resource.

Q168. What is the primary purpose of HTTP 200?

A. Indicates a successful request response.

B. Indicates a client-side bad request.

C. Indicates a generic internal server error.

D. Apple's framework API for HTTP networking and data transfer.

Answer: A. Indicates a successful request response.

Q169. Which description best matches HTTP 201?

A. Stores request URL, method, headers and optional body.

B. Indicates missing or invalid authentication credentials.

C. Indicates a resource was created successfully.

D. Declares the media type of a JSON request or response body.

Answer: C. Indicates a resource was created successfully.

Q170. In iOS development, what does HTTP 204 mean?

A. Carries a bearer access token in an HTTP authorization header.

B. Indicates success without a response body.

C. Retrieves a resource without intending to modify server state.

D. Indicates the server understood the request but refuses authorization.

Answer: B. Indicates success without a response body.

Q171. Choose the accurate description of HTTP 400.

A. Submits data for server-side processing or resource creation.

B. Indicates the requested resource was not found.

C. Safely assembles URLs and percent-encoded query items.

D. Indicates a client-side bad request.

Answer: D. Indicates a client-side bad request.

Q172. What is the primary purpose of HTTP 401?

A. Performs a URL request and returns data plus response asynchronously.

B. Indicates a client has sent too many requests.

C. Indicates missing or invalid authentication credentials.

D. Typically replaces the representation of a specified resource.

Answer: C. Indicates missing or invalid authentication credentials.

Q173. Which description best matches HTTP 403?

A. Apple's framework API for HTTP networking and data transfer.

B. Applies partial modifications to a resource.

C. Indicates the server understood the request but refuses authorization.

D. Indicates a generic internal server error.

Answer: C. Indicates the server understood the request but refuses authorization.

Q174. In iOS development, what does HTTP 404 mean?

A. Declares the media type of a JSON request or response body.

B. Stores request URL, method, headers and optional body.

C. Requests removal of a resource.

D. Indicates the requested resource was not found.

Answer: D. Indicates the requested resource was not found.

Q175. Choose the accurate description of HTTP 429.

A. Indicates a client has sent too many requests.

B. Indicates a successful request response.

C. Carries a bearer access token in an HTTP authorization header.

D. Retrieves a resource without intending to modify server state.

Answer: A. Indicates a client has sent too many requests.

Q176. What is the primary purpose of HTTP 500?

A. Submits data for server-side processing or resource creation.

B. Indicates a generic internal server error.

C. Indicates a resource was created successfully.

D. Safely assembles URLs and percent-encoded query items.

Answer: B. Indicates a generic internal server error.

Q177. Which description best matches Content-Type: application/json?

A. Declares the media type of a JSON request or response body.

B. Indicates success without a response body.

C. Performs a URL request and returns data plus response asynchronously.

D. Typically replaces the representation of a specified resource.

Answer: A. Declares the media type of a JSON request or response body.

Q178. In iOS development, what does Authorization: Bearer mean?

A. Carries a bearer access token in an HTTP authorization header.

B. Applies partial modifications to a resource.

C. Apple's framework API for HTTP networking and data transfer.

D. Indicates a client-side bad request.

Answer: A. Carries a bearer access token in an HTTP authorization header.

Q179. Choose the accurate description of URLComponents.

A. Stores request URL, method, headers and optional body.

B. Requests removal of a resource.

C. Indicates missing or invalid authentication credentials.

D. Safely assembles URLs and percent-encoded query items.

Answer: D. Safely assembles URLs and percent-encoded query items.

Q180. What is the primary purpose of URLSession.data(for:)?

A. Retrieves a resource without intending to modify server state.

B. Indicates a successful request response.

C. Performs a URL request and returns data plus response asynchronously.

D. Indicates the server understood the request but refuses authorization.

Answer: C. Performs a URL request and returns data plus response asynchronously.

10 JSON, Persistence and Security

Q181. Which description best matches JSONDecoder?

A. Converts many snake_case keys to camelCase property names.
B. Stores recreatable data that the system may remove.
C. Tracks managed objects and coordinates changes before persistence.
D. Converts JSON data into Decodable Swift values.

Answer: D. Converts JSON data into Decodable Swift values.

Q182. In iOS development, what does JSONEncoder mean?

A. Decodes ISO 8601 date strings with supported formatter behavior.
B. Caches eligible URL responses based on caching policy.
C. Converts Encodable Swift values into JSON data.
D. Sets up and encapsulates a Core Data persistence stack.

Answer: C. Converts Encodable Swift values into JSON data.

Q183. Choose the accurate description of CodingKeys.

A. Provides an embedded relational database engine.
B. Maps Swift property names to serialized coding keys.
C. Enforces secure network transport policies by default.
D. Stores small preferences and settings rather than secrets.

Answer: B. Maps Swift property names to serialized coding keys.

Q184. What is the primary purpose of keyDecodingStrategy.convertFromSnakeCase?

A. Manages files and directories in the app sandbox.
B. Stores small sensitive credentials using system-protected storage.
C. Protects traffic in transit through TLS encryption.
D. Converts many snake_case keys to camelCase property names.

Answer: D. Converts many snake_case keys to camelCase property names.

Q185. Which description best matches DateDecodingStrategy.iso8601?

A. Adds verification constraints on acceptable server certificates or public keys.
B. Decodes ISO 8601 date strings with supported formatter behavior.
C. Provides an object-graph management and persistence framework.
D. Stores user-generated files that should generally be backed up.

Answer: B. Decodes ISO 8601 date strings with supported formatter behavior.

Q186. In iOS development, what does UserDefaults mean?

A. Stores recreatable data that the system may remove.
B. Stores small preferences and settings rather than secrets.
C. Provides hardware-backed protection for supported cryptographic key operations.
D. Tracks managed objects and coordinates changes before persistence.

Answer: B. Stores small preferences and settings rather than secrets.

Q187. Choose the accurate description of Keychain.

A. Provides Swift APIs for hashing, signing and cryptographic operations.
B. Sets up and encapsulates a Core Data persistence stack.
C. Caches eligible URL responses based on caching policy.
D. Stores small sensitive credentials using system-protected storage.

Answer: D. Stores small sensitive credentials using system-protected storage.

Q188. What is the primary purpose of Core Data?

A. Provides an object-graph management and persistence framework.
B. Enforces secure network transport policies by default.
C. Converts JSON data into Decodable Swift values.
D. Provides an embedded relational database engine.

Answer: A. Provides an object-graph management and persistence framework.

Q189. Which description best matches NSManagedObjectContext?

A. Converts Encodable Swift values into JSON data.
B. Protects traffic in transit through TLS encryption.
C. Manages files and directories in the app sandbox.
D. Tracks managed objects and coordinates changes before persistence.

Answer: D. Tracks managed objects and coordinates changes before persistence.

Q190. In iOS development, what does NSPersistentContainer mean?

A. Adds verification constraints on acceptable server certificates or public keys.
B. Maps Swift property names to serialized coding keys.
C. Sets up and encapsulates a Core Data persistence stack.
D. Stores user-generated files that should generally be backed up.

Answer: C. Sets up and encapsulates a Core Data persistence stack.

Q191. Choose the accurate description of SQLite.

A. Provides an embedded relational database engine.
B. Stores recreatable data that the system may remove.
C. Converts many snake_case keys to camelCase property names.
D. Provides hardware-backed protection for supported cryptographic key operations.

Answer: A. Provides an embedded relational database engine.

Q192. What is the primary purpose of FileManager?

A. Decodes ISO 8601 date strings with supported formatter behavior.
B. Caches eligible URL responses based on caching policy.
C. Provides Swift APIs for hashing, signing and cryptographic operations.
D. Manages files and directories in the app sandbox.

Answer: D. Manages files and directories in the app sandbox.

Q193. Which description best matches Documents directory?

A. Converts JSON data into Decodable Swift values.
B. Stores user-generated files that should generally be backed up.
C. Stores small preferences and settings rather than secrets.
D. Enforces secure network transport policies by default.

Answer: B. Stores user-generated files that should generally be backed up.

Q194. In iOS development, what does Caches directory mean?

A. Protects traffic in transit through TLS encryption.
B. Stores small sensitive credentials using system-protected storage.
C. Converts Encodable Swift values into JSON data.
D. Stores recreatable data that the system may remove.

Answer: D. Stores recreatable data that the system may remove.

Q195. Choose the accurate description of URLCache.

A. Provides an object-graph management and persistence framework.
B. Maps Swift property names to serialized coding keys.
C. Adds verification constraints on acceptable server certificates or public keys.
D. Caches eligible URL responses based on caching policy.

Answer: D. Caches eligible URL responses based on caching policy.

Q196. What is the primary purpose of App Transport Security?

A. Enforces secure network transport policies by default.
B. Tracks managed objects and coordinates changes before persistence.
C. Provides hardware-backed protection for supported cryptographic key operations.
D. Converts many snake_case keys to camelCase property names.

Answer: A. Enforces secure network transport policies by default.

Q197. Which description best matches HTTPS?

A. Protects traffic in transit through TLS encryption.
B. Decodes ISO 8601 date strings with supported formatter behavior.
C. Sets up and encapsulates a Core Data persistence stack.
D. Provides Swift APIs for hashing, signing and cryptographic operations.

Answer: A. Protects traffic in transit through TLS encryption.

Q198. In iOS development, what does certificate pinning mean?

A. Stores small preferences and settings rather than secrets.
B. Provides an embedded relational database engine.
C. Adds verification constraints on acceptable server certificates or public keys.
D. Converts JSON data into Decodable Swift values.

Answer: C. Adds verification constraints on acceptable server certificates or public keys.

Q199. Choose the accurate description of Secure Enclave.

A. Manages files and directories in the app sandbox.

B. Converts Encodable Swift values into JSON data.

C. Stores small sensitive credentials using system-protected storage.

D. Provides hardware-backed protection for supported cryptographic key operations.

Answer: D. Provides hardware-backed protection for supported cryptographic key operations.

Q200. What is the primary purpose of CryptoKit?

A. Stores user-generated files that should generally be backed up.

B. Maps Swift property names to serialized coding keys.

C. Provides Swift APIs for hashing, signing and cryptographic operations.

D. Provides an object-graph management and persistence framework.

Answer: C. Provides Swift APIs for hashing, signing and cryptographic operations.

11 Architecture and Design Patterns

Q201. Which description best matches MVC?

A. Separates model, view and controller responsibilities.
B. Abstracts how domain data is obtained and stored.
C. Moves navigation flow decisions out of individual view controllers.
D. Requires subtypes to preserve the contracts expected from their base types.

Answer: A. Separates model, view and controller responsibilities.

Q202. In iOS development, what does MVVM mean?

A. Separates view rendering from view-model presentation state and behavior.
B. Supplies dependencies externally instead of constructing them within a component.
C. Favors focused interfaces rather than forcing unrelated requirements.
D. Notifies subscribers when observed state or events change.

Answer: A. Separates view rendering from view-model presentation state and behavior.

Q203. Choose the accurate description of ViewModel.

A. Models behavior through protocol contracts and conformance.
B. Makes high-level code depend on abstractions rather than concrete low-level details.
C. Routes events or decisions to another object via a defined interface.
D. Prepares presentation-ready state and coordinates actions for the view.

Answer: D. Prepares presentation-ready state and coordinates actions for the view.

Q204. What is the primary purpose of Repository pattern?

A. Assigns distinct responsibilities to distinct modules or layers.
B. Ensures one shared instance accessed through a common point.
C. Abstracts how domain data is obtained and stored.
D. Says a component should have one primary reason to change.

Answer: C. Abstracts how domain data is obtained and stored.

Q205. Which description best matches Dependency injection?

A. Allows fake transport or service implementations in unit tests.
B. Centralizes construction of objects without forcing callers to know concrete creation details.
C. Supplies dependencies externally instead of constructing them within a component.
D. Favors extension of behavior without modifying stable existing code.

Answer: C. Supplies dependencies externally instead of constructing them within a component.

Q206. In iOS development, what does protocol-oriented design mean?

A. Models behavior through protocol contracts and conformance.
B. Requires subtypes to preserve the contracts expected from their base types.
C. Moves state changes through predictable event-action-update paths.
D. Moves navigation flow decisions out of individual view controllers.

Answer: A. Models behavior through protocol contracts and conformance.

Q207. Choose the accurate description of Singleton.

A. Ensures one shared instance accessed through a common point.

B. Favors focused interfaces rather than forcing unrelated requirements.

C. Splits an application into independently maintainable feature boundaries.

D. Notifies subscribers when observed state or events change.

Answer: A. Ensures one shared instance accessed through a common point.

Q208. What is the primary purpose of Factory pattern?

A. Centralizes construction of objects without forcing callers to know concrete creation details.

B. Separates model, view and controller responsibilities.

C. Makes high-level code depend on abstractions rather than concrete low-level details.

D. Routes events or decisions to another object via a defined interface.

Answer: A. Centralizes construction of objects without forcing callers to know concrete creation details.

Q209. Which description best matches Coordinator pattern?

A. Says a component should have one primary reason to change.

B. Assigns distinct responsibilities to distinct modules or layers.

C. Separates view rendering from view-model presentation state and behavior.

D. Moves navigation flow decisions out of individual view controllers.

Answer: D. Moves navigation flow decisions out of individual view controllers.

Q210. In iOS development, what does Observer pattern mean?

A. Prepares presentation-ready state and coordinates actions for the view.

B. Notifies subscribers when observed state or events change.

C. Favors extension of behavior without modifying stable existing code.

D. Allows fake transport or service implementations in unit tests.

Answer: B. Notifies subscribers when observed state or events change.

Q211. Choose the accurate description of Delegate pattern.

A. Routes events or decisions to another object via a defined interface.

B. Requires subtypes to preserve the contracts expected from their base types.

C. Moves state changes through predictable event-action-update paths.

D. Abstracts how domain data is obtained and stored.

Answer: A. Routes events or decisions to another object via a defined interface.

Q212. What is the primary purpose of SOLID SRP?

A. Says a component should have one primary reason to change.

B. Favors focused interfaces rather than forcing unrelated requirements.

C. Supplies dependencies externally instead of constructing them within a component.

D. Splits an application into independently maintainable feature boundaries.

Answer: A. Says a component should have one primary reason to change.

Q213. Which description best matches SOLID OCP?

A. Separates model, view and controller responsibilities.

B. Favors extension of behavior without modifying stable existing code.

C. Models behavior through protocol contracts and conformance.

D. Makes high-level code depend on abstractions rather than concrete low-level details.

Answer: B. Favors extension of behavior without modifying stable existing code.

Q214. In iOS development, what does SOLID LSP mean?

A. Ensures one shared instance accessed through a common point.

B. Separates view rendering from view-model presentation state and behavior.

C. Requires subtypes to preserve the contracts expected from their base types.

D. Assigns distinct responsibilities to distinct modules or layers.

Answer: C. Requires subtypes to preserve the contracts expected from their base types.

Q215. Choose the accurate description of SOLID ISP.

A. Centralizes construction of objects without forcing callers to know concrete creation details.

B. Allows fake transport or service implementations in unit tests.

C. Prepares presentation-ready state and coordinates actions for the view.

D. Favors focused interfaces rather than forcing unrelated requirements.

Answer: D. Favors focused interfaces rather than forcing unrelated requirements.

Q216. What is the primary purpose of SOLID DIP?

A. Moves navigation flow decisions out of individual view controllers.

B. Makes high-level code depend on abstractions rather than concrete low-level details.

C. Moves state changes through predictable event-action-update paths.

D. Abstracts how domain data is obtained and stored.

Answer: B. Makes high-level code depend on abstractions rather than concrete low-level details.

Q217. Which description best matches Separation of concerns?

A. Supplies dependencies externally instead of constructing them within a component.

B. Assigns distinct responsibilities to distinct modules or layers.

C. Splits an application into independently maintainable feature boundaries.

D. Notifies subscribers when observed state or events change.

Answer: B. Assigns distinct responsibilities to distinct modules or layers.

Q218. In iOS development, what does Testable networking abstraction mean?

A. Separates model, view and controller responsibilities.

B. Routes events or decisions to another object via a defined interface.

C. Allows fake transport or service implementations in unit tests.

D. Models behavior through protocol contracts and conformance.

Answer: C. Allows fake transport or service implementations in unit tests.

Q219. Choose the accurate description of Unidirectional data flow.

A. Moves state changes through predictable event-action-update paths.

B. Says a component should have one primary reason to change.

C. Ensures one shared instance accessed through a common point.

D. Separates view rendering from view-model presentation state and behavior.

Answer: A. Moves state changes through predictable event-action-update paths.

Q220. What is the primary purpose of Feature modularization?

A. Favors extension of behavior without modifying stable existing code.

B. Centralizes construction of objects without forcing callers to know concrete creation details.

C. Splits an application into independently maintainable feature boundaries.

D. Prepares presentation-ready state and coordinates actions for the view.

Answer: C. Splits an application into independently maintainable feature boundaries.

12 Testing, Debugging and Tooling

Q221. Which description best matches XCTest?

A. Checks that an expression throws an error.
B. Samples CPU usage to identify expensive execution paths.
C. Checks visible flows and interactions through the app interface.
D. Apple's testing framework for unit and performance tests.

Answer: D. Apple's testing framework for unit and performance tests.

Q222. In iOS development, what does XCUITest mean?

A. Diagnoses many UIKit calls made from incorrect threads.
B. Measures code performance such as runtime or memory use.
C. Automates UI-level interactions and assertions.
D. Simulates a collaborator to control test behavior.

Answer: C. Automates UI-level interactions and assertions.

Q223. Choose the accurate description of XCTAssertEqual.

A. Simulates constrained network conditions for testing.
B. Asserts that two values compare equal.
C. Pauses execution at a selected point for inspection.
D. Returns predetermined responses for test inputs.

Answer: B. Asserts that two values compare equal.

Q224. What is the primary purpose of XCTAssertThrowsError?

A. Checks that an expression throws an error.
B. Distributes prerelease builds to internal or external testers.
C. Prints a debug description of an evaluated expression.
D. Checks a small isolated unit of behavior.

Answer: A. Checks that an expression throws an error.

Q225. Which description best matches mock object?

A. Stops a debugger when relevant exceptions are thrown.
B. Checks how multiple parts work together.
C. Simulates a collaborator to control test behavior.
D. Defines build, run, test, profile, analyze and archive actions.

Answer: C. Simulates a collaborator to control test behavior.

Q226. In iOS development, what does stub mean?

A. Samples CPU usage to identify expensive execution paths.
B. Returns predetermined responses for test inputs.
C. Stores reusable build settings in a configuration file.
D. Checks visible flows and interactions through the app interface.

Answer: B. Returns predetermined responses for test inputs.

Q227. Choose the accurate description of unit test.

A. Checks a small isolated unit of behavior.
B. Manages Swift package dependencies and builds.
C. Measures code performance such as runtime or memory use.
D. Diagnoses many UIKit calls made from incorrect threads.

Answer: A. Checks a small isolated unit of behavior.

Q228. What is the primary purpose of integration test?

A. Apple's testing framework for unit and performance tests.
B. Checks how multiple parts work together.
C. Simulates constrained network conditions for testing.
D. Pauses execution at a selected point for inspection.

Answer: B. Checks how multiple parts work together.

Q229. Which description best matches UI test?

A. Distributes prerelease builds to internal or external testers.
B. Checks visible flows and interactions through the app interface.
C. Prints a debug description of an evaluated expression.
D. Automates UI-level interactions and assertions.

Answer: B. Checks visible flows and interactions through the app interface.

Q230. In iOS development, what does performance test mean?

A. Asserts that two values compare equal.
B. Defines build, run, test, profile, analyze and archive actions.
C. Stops a debugger when relevant exceptions are thrown.
D. Measures code performance such as runtime or memory use.

Answer: D. Measures code performance such as runtime or memory use.

Q231. Choose the accurate description of breakpoint.

A. Checks that an expression throws an error.
B. Samples CPU usage to identify expensive execution paths.
C. Stores reusable build settings in a configuration file.
D. Pauses execution at a selected point for inspection.

Answer: D. Pauses execution at a selected point for inspection.

Q232. What is the primary purpose of LLDB po?

A. Prints a debug description of an evaluated expression.
B. Diagnoses many UIKit calls made from incorrect threads.
C. Manages Swift package dependencies and builds.
D. Simulates a collaborator to control test behavior.

Answer: A. Prints a debug description of an evaluated expression.

Q233. Which description best matches exception breakpoint?

A. Stops a debugger when relevant exceptions are thrown.

B. Apple's testing framework for unit and performance tests.

C. Returns predetermined responses for test inputs.

D. Simulates constrained network conditions for testing.

Answer: A. Stops a debugger when relevant exceptions are thrown.

Q234. In iOS development, what does Instruments Time Profiler mean?

A. Automates UI-level interactions and assertions.

B. Checks a small isolated unit of behavior.

C. Distributes prerelease builds to internal or external testers.

D. Samples CPU usage to identify expensive execution paths.

Answer: D. Samples CPU usage to identify expensive execution paths.

Q235. Choose the accurate description of Main Thread Checker.

A. Checks how multiple parts work together.

B. Defines build, run, test, profile, analyze and archive actions.

C. Asserts that two values compare equal.

D. Diagnoses many UIKit calls made from incorrect threads.

Answer: D. Diagnoses many UIKit calls made from incorrect threads.

Q236. What is the primary purpose of Network Link Conditioner?

A. Checks that an expression throws an error.

B. Stores reusable build settings in a configuration file.

C. Checks visible flows and interactions through the app interface.

D. Simulates constrained network conditions for testing.

Answer: D. Simulates constrained network conditions for testing.

Q237. Which description best matches TestFlight?

A. Measures code performance such as runtime or memory use.

B. Manages Swift package dependencies and builds.

C. Distributes prerelease builds to internal or external testers.

D. Simulates a collaborator to control test behavior.

Answer: C. Distributes prerelease builds to internal or external testers.

Q238. In iOS development, what does Xcode scheme mean?

A. Returns predetermined responses for test inputs.

B. Pauses execution at a selected point for inspection.

C. Defines build, run, test, profile, analyze and archive actions.

D. Apple's testing framework for unit and performance tests.

Answer: C. Defines build, run, test, profile, analyze and archive actions.

Q239. Choose the accurate description of xcconfig.

A. Checks a small isolated unit of behavior.

B. Automates UI-level interactions and assertions.

C. Prints a debug description of an evaluated expression.

D. Stores reusable build settings in a configuration file.

Answer: D. Stores reusable build settings in a configuration file.

Q240. What is the primary purpose of Swift Package Manager?

A. Stops a debugger when relevant exceptions are thrown.

B. Manages Swift package dependencies and builds.

C. Asserts that two values compare equal.

D. Checks how multiple parts work together.

Answer: B. Manages Swift package dependencies and builds.

13 App Lifecycle and Apple Frameworks

Q241. Which description best matches application(_:didFinishLaunchingWithOptions:)?

A. Identifies a device-app installation for Apple Push Notification service routing.
B. Provides APIs for audio, video, camera capture and media processing.
C. Called as a scene transitions away from active state.
D. Application delegate callback for launch initialization.

Answer: D. Application delegate callback for launch initialization.

Q242. In iOS development, what does scene(_:willConnectTo:options:) mean?

A. Opens an app to a specific destination using a URL or universal link.
B. Called when a scene enters the background.
C. Scene delegate callback when a UI scene connects.
D. Plays timed audiovisual media content.

Answer: C. Scene delegate callback when a UI scene connects.

Q243. Choose the accurate description of sceneDidBecomeActive.

A. Provides layer-based animation and compositing infrastructure.
B. Uses an associated HTTPS web URL to route eligible links into an installed app.
C. Legacy application lifecycle callback for entering background.
D. Called when a scene becomes active for user interaction.

Answer: D. Called when a scene becomes active for user interaction.

Q244. What is the primary purpose of sceneWillResignActive?

A. Called as a scene transitions away from active state.
B. Declares domains used for universal links and related features.
C. Provides low-level 2D drawing APIs.
D. Schedules supported background processing or refresh opportunities.

Answer: A. Called as a scene transitions away from active state.

Q245. Which description best matches sceneDidEnterBackground?

A. Provides location and heading services with required permissions.
B. Called when a scene enters the background.
C. Coordinates notification authorization, scheduling and delivery.
D. Builds widgets that show glanceable information outside the app.

Answer: B. Called when a scene enters the background.

Q246. In iOS development, what does applicationDidEnterBackground mean?

A. Legacy application lifecycle callback for entering background.
B. Identifies a device-app installation for Apple Push Notification service routing.
C. Provides APIs for audio, video, camera capture and media processing.
D. Enables supported data sharing between related app targets.

Answer: A. Legacy application lifecycle callback for entering background.

Q247. Choose the accurate description of BGTaskScheduler.

A. Schedules supported background processing or refresh opportunities.

B. Explains to users why the app requests privacy-sensitive access.

C. Plays timed audiovisual media content.

D. Opens an app to a specific destination using a URL or universal link.

Answer: A. Schedules supported background processing or refresh opportunities.

Q248. What is the primary purpose of UNUserNotificationCenter?

A. Application delegate callback for launch initialization.

B. Uses an associated HTTPS web URL to route eligible links into an installed app.

C. Provides layer-based animation and compositing infrastructure.

D. Coordinates notification authorization, scheduling and delivery.

Answer: D. Coordinates notification authorization, scheduling and delivery.

Q249. Which description best matches APNs device token?

A. Identifies a device-app installation for Apple Push Notification service routing.

B. Declares domains used for universal links and related features.

C. Scene delegate callback when a UI scene connects.

D. Provides low-level 2D drawing APIs.

Answer: A. Identifies a device-app installation for Apple Push Notification service routing.

Q250. In iOS development, what does deep link mean?

A. Called when a scene becomes active for user interaction.

B. Provides location and heading services with required permissions.

C. Builds widgets that show glanceable information outside the app.

D. Opens an app to a specific destination using a URL or universal link.

Answer: D. Opens an app to a specific destination using a URL or universal link.

Q251. Choose the accurate description of universal link.

A. Enables supported data sharing between related app targets.

B. Provides APIs for audio, video, camera capture and media processing.

C. Called as a scene transitions away from active state.

D. Uses an associated HTTPS web URL to route eligible links into an installed app.

Answer: D. Uses an associated HTTPS web URL to route eligible links into an installed app.

Q252. What is the primary purpose of Associated Domains entitlement?

A. Declares domains used for universal links and related features.

B. Explains to users why the app requests privacy-sensitive access.

C. Plays timed audiovisual media content.

D. Called when a scene enters the background.

Answer: A. Declares domains used for universal links and related features.

Q253. Which description best matches Core Location?

A. Provides layer-based animation and compositing infrastructure.
B. Application delegate callback for launch initialization.
C. Legacy application lifecycle callback for entering background.
D. Provides location and heading services with required permissions.

Answer: D. Provides location and heading services with required permissions.

Q254. In iOS development, what does AVFoundation mean?

A. Provides APIs for audio, video, camera capture and media processing.
B. Schedules supported background processing or refresh opportunities.
C. Provides low-level 2D drawing APIs.
D. Scene delegate callback when a UI scene connects.

Answer: A. Provides APIs for audio, video, camera capture and media processing.

Q255. Choose the accurate description of AVPlayer.

A. Builds widgets that show glanceable information outside the app.
B. Plays timed audiovisual media content.
C. Coordinates notification authorization, scheduling and delivery.
D. Called when a scene becomes active for user interaction.

Answer: B. Plays timed audiovisual media content.

Q256. What is the primary purpose of Core Animation?

A. Called as a scene transitions away from active state.
B. Identifies a device-app installation for Apple Push Notification service routing.
C. Provides layer-based animation and compositing infrastructure.
D. Enables supported data sharing between related app targets.

Answer: C. Provides layer-based animation and compositing infrastructure.

Q257. Which description best matches Core Graphics?

A. Opens an app to a specific destination using a URL or universal link.
B. Called when a scene enters the background.
C. Provides low-level 2D drawing APIs.
D. Explains to users why the app requests privacy-sensitive access.

Answer: C. Provides low-level 2D drawing APIs.

Q258. In iOS development, what does WidgetKit mean?

A. Uses an associated HTTPS web URL to route eligible links into an installed app.
B. Legacy application lifecycle callback for entering background.
C. Builds widgets that show glanceable information outside the app.
D. Application delegate callback for launch initialization.

Answer: C. Builds widgets that show glanceable information outside the app.

Q259. Choose the accurate description of App Groups.

A. Declares domains used for universal links and related features.

B. Enables supported data sharing between related app targets.

C. Schedules supported background processing or refresh opportunities.

D. Scene delegate callback when a UI scene connects.

Answer: B. Enables supported data sharing between related app targets.

Q260. What is the primary purpose of Info.plist usage description key?

A. Explains to users why the app requests privacy-sensitive access.

B. Provides location and heading services with required permissions.

C. Coordinates notification authorization, scheduling and delivery.

D. Called when a scene becomes active for user interaction.

Answer: A. Explains to users why the app requests privacy-sensitive access.

14 Combine and Reactive Programming

Q261. Which description best matches Publisher?

A. Emits a sequence of values over time and possibly a completion.
B. Attaches a subscriber using value and optional completion closures.
C. Suppresses consecutive equal elements.
D. Transforms emitted values into a new publisher of mapped values.

Answer: A. Emits a sequence of values over time and possibly a completion.

Q262. In iOS development, what does Subscriber mean?

A. Forwards received publisher values into an object property.
B. Combines the latest emissions of multiple publishers once each has emitted.
C. Receives values and completion from a publisher after subscription.
D. Forwards only emitted values meeting a predicate.

Answer: C. Receives values and completion from a publisher after subscription.

Q263. Choose the accurate description of AnyCancellable.

A. Broadcasts manually sent values without storing a current value.
B. Transforms emitted values while dropping nil outputs.
C. Represents a cancellable subscription and cancels when released.
D. Pairs corresponding elements from multiple publishers.

Answer: C. Represents a cancellable subscription and cancels when released.

Q264. What is the primary purpose of sink?

A. Attaches a subscriber using value and optional completion closures.
B. Stores and publishes a current value to new subscribers.
C. Waits for a quiet interval before forwarding a recent value.
D. Transforms each emitted value into a publisher and merges output.

Answer: A. Attaches a subscriber using value and optional completion closures.

Q265. Which description best matches assign?

A. Limits emissions to a specified time interval.
B. Changes the scheduler used to deliver downstream values.
C. Forwards received publisher values into an object property.
D. Emits values when a published property changes.

Answer: C. Forwards received publisher values into an object property.

Q266. In iOS development, what does PassthroughSubject mean?

A. Changes the scheduler used for upstream subscription work.
B. Transforms emitted values into a new publisher of mapped values.
C. Suppresses consecutive equal elements.
D. Broadcasts manually sent values without storing a current value.

Answer: D. Broadcasts manually sent values without storing a current value.

Q267. Choose the accurate description of CurrentValueSubject.

A. Forwards only emitted values meeting a predicate.

B. Combines the latest emissions of multiple publishers once each has emitted.

C. Hides a concrete publisher type behind AnyPublisher.

D. Stores and publishes a current value to new subscribers.

Answer: D. Stores and publishes a current value to new subscribers.

Q268. What is the primary purpose of @Published publisher?

A. Emits a sequence of values over time and possibly a completion.

B. Pairs corresponding elements from multiple publishers.

C. Transforms emitted values while dropping nil outputs.

D. Emits values when a published property changes.

Answer: D. Emits values when a published property changes.

Q269. Which description best matches map operator?

A. Receives values and completion from a publisher after subscription.

B. Transforms emitted values into a new publisher of mapped values.

C. Transforms each emitted value into a publisher and merges output.

D. Waits for a quiet interval before forwarding a recent value.

Answer: B. Transforms emitted values into a new publisher of mapped values.

Q270. In iOS development, what does filter operator mean?

A. Forwards only emitted values meeting a predicate.

B. Changes the scheduler used to deliver downstream values.

C. Represents a cancellable subscription and cancels when released.

D. Limits emissions to a specified time interval.

Answer: A. Forwards only emitted values meeting a predicate.

Q271. Choose the accurate description of compactMap operator.

A. Changes the scheduler used for upstream subscription work.

B. Suppresses consecutive equal elements.

C. Transforms emitted values while dropping nil outputs.

D. Attaches a subscriber using value and optional completion closures.

Answer: C. Transforms emitted values while dropping nil outputs.

Q272. What is the primary purpose of debounce?

A. Forwards received publisher values into an object property.

B. Combines the latest emissions of multiple publishers once each has emitted.

C. Waits for a quiet interval before forwarding a recent value.

D. Hides a concrete publisher type behind AnyPublisher.

Answer: C. Waits for a quiet interval before forwarding a recent value.

Q273. Which description best matches throttle?

A. Emits a sequence of values over time and possibly a completion.
B. Broadcasts manually sent values without storing a current value.
C. Limits emissions to a specified time interval.
D. Pairs corresponding elements from multiple publishers.

Answer: C. Limits emissions to a specified time interval.

Q274. In iOS development, what does removeDuplicates mean?

A. Transforms each emitted value into a publisher and merges output.
B. Stores and publishes a current value to new subscribers.
C. Receives values and completion from a publisher after subscription.
D. Suppresses consecutive equal elements.

Answer: D. Suppresses consecutive equal elements.

Q275. Choose the accurate description of combineLatest.

A. Emits values when a published property changes.
B. Represents a cancellable subscription and cancels when released.
C. Combines the latest emissions of multiple publishers once each has emitted.
D. Changes the scheduler used to deliver downstream values.

Answer: C. Combines the latest emissions of multiple publishers once each has emitted.

Q276. What is the primary purpose of zip?

A. Attaches a subscriber using value and optional completion closures.
B. Transforms emitted values into a new publisher of mapped values.
C. Pairs corresponding elements from multiple publishers.
D. Changes the scheduler used for upstream subscription work.

Answer: C. Pairs corresponding elements from multiple publishers.

Q277. Which description best matches flatMap operator?

A. Transforms each emitted value into a publisher and merges output.
B. Forwards only emitted values meeting a predicate.
C. Hides a concrete publisher type behind AnyPublisher.
D. Forwards received publisher values into an object property.

Answer: A. Transforms each emitted value into a publisher and merges output.

Q278. In iOS development, what does receive(on:) mean?

A. Broadcasts manually sent values without storing a current value.
B. Changes the scheduler used to deliver downstream values.
C. Transforms emitted values while dropping nil outputs.
D. Emits a sequence of values over time and possibly a completion.

Answer: B. Changes the scheduler used to deliver downstream values.

Q279. Choose the accurate description of subscribe(on:).

A. Changes the scheduler used for upstream subscription work.

B. Waits for a quiet interval before forwarding a recent value.

C. Receives values and completion from a publisher after subscription.

D. Stores and publishes a current value to new subscribers.

Answer: A. Changes the scheduler used for upstream subscription work.

Q280. What is the primary purpose of .eraseToAnyPublisher()?

A. Limits emissions to a specified time interval.

B. Emits values when a published property changes.

C. Represents a cancellable subscription and cancels when released.

D. Hides a concrete publisher type behind AnyPublisher.

Answer: D. Hides a concrete publisher type behind AnyPublisher.

15 Practical iOS Scenarios

Q281. Which description best matches Search field API flood?

A. Debounce user text changes before making search requests.
B. Verify permission, token registration, entitlement, environment and server payload.
C. Keep a persistent local source of truth and synchronize when online.
D. Store sensitive access credentials in Keychain rather than UserDefaults.

Answer: A. Debounce user text changes before making search requests.

Q282. In iOS development, what does Offscreen image memory spike mean?

A. Replace unsafe force unwrap with guarded optional handling.
B. Downsample oversized images and configure caching appropriately.
C. Inspect strong references, observers, timers and closure captures.
D. Handle HTTP 401 and use an appropriate token refresh or reauthentication flow.

Answer: B. Downsample oversized images and configure caching appropriately.

Q283. Choose the accurate description of Reused cell wrong avatar.

A. Tie work to stable task identity and avoid fetching during body evaluation.
B. Respect HTTP 429 and Retry-After guidance with bounded backoff.
C. Cancel or identify in-flight image requests before applying results to reused cells.
D. Perform expensive parsing away from the main actor and update UI on main.

Answer: C. Cancel or identify in-flight image requests before applying results to reused cells.

Q284. What is the primary purpose of Login token persistence?

A. Avoid synchronous image decoding and expensive cell configuration on the main thread.
B. Ensure presentation occurs from a valid active presenter on the main thread.
C. Add sufficient non-conflicting positional and sizing constraints.
D. Store sensitive access credentials in Keychain rather than UserDefaults.

Answer: D. Store sensitive access credentials in Keychain rather than UserDefaults.

Q285. Which description best matches View controller never deallocates?

A. Route the incoming URL after app and navigation state are ready.
B. Inject a mock HTTP client that returns controlled failures.
C. Cancel previous work or check request identity before applying results.
D. Inspect strong references, observers, timers and closure captures.

Answer: D. Inspect strong references, observers, timers and closure captures.

Q286. In iOS development, what does UI freezes after JSON parsing mean?

A. Keep a persistent local source of truth and synchronize when online.
B. Perform expensive parsing away from the main actor and update UI on main.
C. Move true secrets to a secure server rather than relying on client obfuscation.
D. Verify permission, token registration, entitlement, environment and server payload.

Answer: B. Perform expensive parsing away from the main actor and update UI on main.

Q287. Choose the accurate description of Slow scrolling with images.

A. Avoid synchronous image decoding and expensive cell configuration on the main thread.

B. Use XCTest async support or expectations rather than arbitrary sleeps.

C. Handle HTTP 401 and use an appropriate token refresh or reauthentication flow.

D. Replace unsafe force unwrap with guarded optional handling.

Answer: A. Avoid synchronous image decoding and expensive cell configuration on the main thread.

Q288. What is the primary purpose of Stale network response overwrites search?

A. Cancel previous work or check request identity before applying results.

B. Debounce user text changes before making search requests.

C. Respect HTTP 429 and Retry-After guidance with bounded backoff.

D. Tie work to stable task identity and avoid fetching during body evaluation.

Answer: A. Cancel previous work or check request identity before applying results.

Q289. Which description best matches Offline saved favorites?

A. Add sufficient non-conflicting positional and sizing constraints.

B. Keep a persistent local source of truth and synchronize when online.

C. Ensure presentation occurs from a valid active presenter on the main thread.

D. Downsample oversized images and configure caching appropriately.

Answer: B. Keep a persistent local source of truth and synchronize when online.

Q290. In iOS development, what does Unauthorized API response mean?

A. Handle HTTP 401 and use an appropriate token refresh or reauthentication flow.

B. Inject a mock HTTP client that returns controlled failures.

C. Cancel or identify in-flight image requests before applying results to reused cells.

D. Route the incoming URL after app and navigation state are ready.

Answer: A. Handle HTTP 401 and use an appropriate token refresh or reauthentication flow.

Q291. Choose the accurate description of Rate-limited API response.

A. Respect HTTP 429 and Retry-After guidance with bounded backoff.

B. Verify permission, token registration, entitlement, environment and server payload.

C. Move true secrets to a secure server rather than relying on client obfuscation.

D. Store sensitive access credentials in Keychain rather than UserDefaults.

Answer: A. Respect HTTP 429 and Retry-After guidance with bounded backoff.

Q292. What is the primary purpose of Modal never appears?

A. Ensure presentation occurs from a valid active presenter on the main thread.

B. Replace unsafe force unwrap with guarded optional handling.

C. Inspect strong references, observers, timers and closure captures.

D. Use XCTest async support or expectations rather than arbitrary sleeps.

Answer: A. Ensure presentation occurs from a valid active presenter on the main thread.

Q293. Which description best matches Deep link after cold launch?

A. Tie work to stable task identity and avoid fetching during body evaluation.

B. Debounce user text changes before making search requests.

C. Perform expensive parsing away from the main actor and update UI on main.

D. Route the incoming URL after app and navigation state are ready.

Answer: D. Route the incoming URL after app and navigation state are ready.

Q294. In iOS development, what does Push notifications not arriving mean?

A. Verify permission, token registration, entitlement, environment and server payload.

B. Avoid synchronous image decoding and expensive cell configuration on the main thread.

C. Downsample oversized images and configure caching appropriately.

D. Add sufficient non-conflicting positional and sizing constraints.

Answer: A. Verify permission, token registration, entitlement, environment and server payload.

Q295. Choose the accurate description of Crash on forced optional unwrap.

A. Inject a mock HTTP client that returns controlled failures.

B. Cancel or identify in-flight image requests before applying results to reused cells.

C. Replace unsafe force unwrap with guarded optional handling.

D. Cancel previous work or check request identity before applying results.

Answer: C. Replace unsafe force unwrap with guarded optional handling.

Q296. What is the primary purpose of Duplicate network fetch in SwiftUI?

A. Move true secrets to a secure server rather than relying on client obfuscation.

B. Keep a persistent local source of truth and synchronize when online.

C. Store sensitive access credentials in Keychain rather than UserDefaults.

D. Tie work to stable task identity and avoid fetching during body evaluation.

Answer: D. Tie work to stable task identity and avoid fetching during body evaluation.

Q297. Which description best matches Autolayout ambiguous warning?

A. Add sufficient non-conflicting positional and sizing constraints.

B. Inspect strong references, observers, timers and closure captures.

C. Use XCTest async support or expectations rather than arbitrary sleeps.

D. Handle HTTP 401 and use an appropriate token refresh or reauthentication flow.

Answer: A. Add sufficient non-conflicting positional and sizing constraints.

Q298. In iOS development, what does Unit testing API errors mean?

A. Respect HTTP 429 and Retry-After guidance with bounded backoff.

B. Debounce user text changes before making search requests.

C. Perform expensive parsing away from the main actor and update UI on main.

D. Inject a mock HTTP client that returns controlled failures.

Answer: D. Inject a mock HTTP client that returns controlled failures.

Q299. Choose the accurate description of App secret embedded in code.

A. Avoid synchronous image decoding and expensive cell configuration on the main thread.

B. Move true secrets to a secure server rather than relying on client obfuscation.

C. Downsample oversized images and configure caching appropriately.

D. Ensure presentation occurs from a valid active presenter on the main thread.

Answer: B. Move true secrets to a secure server rather than relying on client obfuscation.

Q300. What is the primary purpose of Flaky asynchronous unit test?

A. Cancel or identify in-flight image requests before applying results to reused cells.

B. Use XCTest async support or expectations rather than arbitrary sleeps.

C. Route the incoming URL after app and navigation state are ready.

D. Cancel previous work or check request identity before applying results.

Answer: B. Use XCTest async support or expectations rather than arbitrary sleeps.

16 Builds, Signing and App Distribution

Q301. Which description best matches Bundle identifier?

A. Uniquely identifies an app target within Apple's platform ecosystem.
B. Manages app metadata, TestFlight testing and release submissions.
C. Connects an app identifier, signing permissions and supported distribution conditions.
D. Commonly uses development settings, debugging symbols and nonproduction services.

Answer: A. Uniquely identifies an app target within Apple's platform ecosystem.

Q302. In iOS development, what does Development certificate mean?

A. Distributes eligible beta builds to configured App Store Connect team members.
B. Commonly enables production-focused optimization and distribution settings.
C. Signs builds used during development on authorized devices.
D. Identifies an app or group of apps for entitlements and provisioning.

Answer: C. Signs builds used during development on authorized devices.

Q303. Choose the accurate description of Distribution certificate.

A. Declares a capability granted to an app through signing and platform configuration.
B. Distributes beta builds to invited outside testers, subject to applicable review.
C. Selects how an Xcode target is built, run, tested, profiled and archived.
D. Signs an app for an appropriate distribution workflow.

Answer: D. Signs an app for an appropriate distribution workflow.

Q304. What is the primary purpose of Provisioning profile?

A. Lets project and target configuration levels combine or override configuration values.
B. Verifies code origin and helps ensure distributed code integrity.
C. Connects an app identifier, signing permissions and supported distribution conditions.
D. Specifies the marketing version shown to users.

Answer: C. Connects an app identifier, signing permissions and supported distribution conditions.

Q305. Which description best matches App ID?

A. Specifies the build version used to distinguish builds.
B. Identifies an app or group of apps for entitlements and provisioning.
C. Packages a release candidate build for validation and distribution.
D. Contains icon image variants used by the app across supported contexts.

Answer: B. Identifies an app or group of apps for entitlements and provisioning.

Q306. In iOS development, what does Entitlement mean?

A. Converts memory addresses from a crash report into function and source locations using matching symbols.
B. Commonly uses development settings, debugging symbols and nonproduction services.
C. Declares a capability granted to an app through signing and platform configuration.
D. Manages app metadata, TestFlight testing and release submissions.

Answer: C. Declares a capability granted to an app through signing and platform configuration.

Q307. Choose the accurate description of Code signing.

A. Stores debug symbols used to interpret optimized crash reports.

B. Commonly enables production-focused optimization and distribution settings.

C. Verifies code origin and helps ensure distributed code integrity.

D. Distributes eligible beta builds to configured App Store Connect team members.

Answer: C. Verifies code origin and helps ensure distributed code integrity.

Q308. What is the primary purpose of Archive action?

A. Distributes beta builds to invited outside testers, subject to applicable review.

B. Selects how an Xcode target is built, run, tested, profiled and archived.

C. Packages a release candidate build for validation and distribution.

D. Uniquely identifies an app target within Apple's platform ecosystem.

Answer: C. Packages a release candidate build for validation and distribution.

Q309. Which description best matches App Store Connect?

A. Manages app metadata, TestFlight testing and release submissions.

B. Signs builds used during development on authorized devices.

C. Lets project and target configuration levels combine or override configuration values.

D. Specifies the marketing version shown to users.

Answer: A. Manages app metadata, TestFlight testing and release submissions.

Q310. In iOS development, what does TestFlight internal testing mean?

A. Specifies the build version used to distinguish builds.

B. Distributes eligible beta builds to configured App Store Connect team members.

C. Contains icon image variants used by the app across supported contexts.

D. Signs an app for an appropriate distribution workflow.

Answer: B. Distributes eligible beta builds to configured App Store Connect team members.

Q311. Choose the accurate description of TestFlight external testing.

A. Distributes beta builds to invited outside testers, subject to applicable review.

B. Commonly uses development settings, debugging symbols and nonproduction services.

C. Converts memory addresses from a crash report into function and source locations using matching symbols.

D. Connects an app identifier, signing permissions and supported distribution conditions.

Answer: A. Distributes beta builds to invited outside testers, subject to applicable review.

Q312. What is the primary purpose of CFBundleShortVersionString?

A. Stores debug symbols used to interpret optimized crash reports.

B. Identifies an app or group of apps for entitlements and provisioning.

C. Specifies the marketing version shown to users.

D. Commonly enables production-focused optimization and distribution settings.

Answer: C. Specifies the marketing version shown to users.

Q313. Which description best matches CFBundleVersion?

A. Uniquely identifies an app target within Apple's platform ecosystem.

B. Selects how an Xcode target is built, run, tested, profiled and archived.

C. Declares a capability granted to an app through signing and platform configuration.

D. Specifies the build version used to distinguish builds.

Answer: D. Specifies the build version used to distinguish builds.

Q314. In iOS development, what does Debug configuration mean?

A. Verifies code origin and helps ensure distributed code integrity.

B. Lets project and target configuration levels combine or override configuration values.

C. Signs builds used during development on authorized devices.

D. Commonly uses development settings, debugging symbols and nonproduction services.

Answer: D. Commonly uses development settings, debugging symbols and nonproduction services.

Q315. Choose the accurate description of Release configuration.

A. Contains icon image variants used by the app across supported contexts.

B. Packages a release candidate build for validation and distribution.

C. Commonly enables production-focused optimization and distribution settings.

D. Signs an app for an appropriate distribution workflow.

Answer: C. Commonly enables production-focused optimization and distribution settings.

Q316. What is the primary purpose of Scheme?

A. Selects how an Xcode target is built, run, tested, profiled and archived.

B. Converts memory addresses from a crash report into function and source locations using matching symbols.

C. Manages app metadata, TestFlight testing and release submissions.

D. Connects an app identifier, signing permissions and supported distribution conditions.

Answer: A. Selects how an Xcode target is built, run, tested, profiled and archived.

Q317. Which description best matches Build setting inheritance?

A. Distributes eligible beta builds to configured App Store Connect team members.

B. Identifies an app or group of apps for entitlements and provisioning.

C. Stores debug symbols used to interpret optimized crash reports.

D. Lets project and target configuration levels combine or override configuration values.

Answer: D. Lets project and target configuration levels combine or override configuration values.

Q318. In iOS development, what does App icon asset mean?

A. Declares a capability granted to an app through signing and platform configuration.

B. Contains icon image variants used by the app across supported contexts.

C. Distributes beta builds to invited outside testers, subject to applicable review.

D. Uniquely identifies an app target within Apple's platform ecosystem.

Answer: B. Contains icon image variants used by the app across supported contexts.

Q319. Choose the accurate description of Crash symbolication.

A. Specifies the marketing version shown to users.

B. Verifies code origin and helps ensure distributed code integrity.

C. Converts memory addresses from a crash report into function and source locations using matching symbols.

D. Signs builds used during development on authorized devices.

Answer: C. Converts memory addresses from a crash report into function and source locations using matching symbols.

Q320. What is the primary purpose of dSYM file?

A. Signs an app for an appropriate distribution workflow.

B. Specifies the build version used to distinguish builds.

C. Stores debug symbols used to interpret optimized crash reports.

D. Packages a release candidate build for validation and distribution.

Answer: C. Stores debug symbols used to interpret optimized crash reports.

Quick Answer Key

01. Swift Fundamentals

1: C 2: A 3: B 4: D 5: C 6: C 7: A 8: C 9: A 10: D

11: B 12: C 13: B 14: C 15: A 16: D 17: D 18: A 19: B 20: C

02. Value Types and OOP

21: D 22: B 23: A 24: C 25: A 26: D 27: C 28: D 29: B 30: D

31: B 32: B 33: A 34: D 35: C 36: D 37: C 38: C 39: D 40: C

03. Closures, Generics and Error Handling

41: B 42: A 43: A 44: A 45: A 46: A 47: D 48: A 49: C 50: A

51: A 52: A 53: A 54: C 55: D 56: B 57: A 58: C 59: B 60: A

04. Memory Management and ARC

61: D 62: A 63: B 64: C 65: C 66: D 67: A 68: C 69: A 70: D

71: B 72: D 73: C 74: C 75: D 76: D 77: A 78: D 79: A 80: C

05. UIKit Fundamentals

81: A 82: B 83: D 84: C 85: B 86: C 87: C 88: B 89: B 90: B

91: D 92: C 93: B 94: D 95: B 96: D 97: D 98: B 99: D 100: B

06. Auto Layout and UI Performance

101: A 102: D 103: D 104: B 105: D 106: C 107: C 108: A 109: C 110: C

111: D 112: D 113: A 114: A 115: A 116: A 117: D 118: C 119: B 120: B

07. SwiftUI Essentials

121: C 122: A 123: C 124: C 125: C 126: C 127: D 128: A 129: D 130: D

131: D 132: A 133: A 134: D 135: B 136: B 137: B 138: C 139: B 140: D

08. Concurrency and Threading

141: A 142: A 143: D 144: C 145: C 146: C 147: A 148: B 149: B 150: C

151: C 152: B 153: C 154: A 155: B 156: B 157: A 158: D 159: A 160: A

09. Networking and REST APIs

161: B 162: D 163: B 164: A 165: B 166: C 167: D 168: A 169: C 170: B

171: D 172: C 173: C 174: D 175: A 176: B 177: A 178: A 179: D 180: C

10. JSON, Persistence and Security

181: D 182: C 183: B 184: D 185: B 186: B 187: D 188: A 189: D 190: C

191: A 192: D 193: B 194: D 195: D 196: A 197: A 198: C 199: D 200: C

11. Architecture and Design Patterns

201: A 202: A 203: D 204: C 205: C 206: A 207: A 208: A 209: D 210: B

211: A 212: A 213: B 214: C 215: D 216: B 217: B 218: C 219: A 220: C

12. Testing, Debugging and Tooling

221: D 222: C 223: B 224: A 225: C 226: B 227: A 228: B 229: B 230: D

231: D 232: A 233: A 234: D 235: D 236: D 237: C 238: C 239: D 240: B

13. App Lifecycle and Apple Frameworks

241: D 242: C 243: D 244: A 245: B 246: A 247: A 248: D 249: A 250: D

251: D 252: A 253: D 254: A 255: B 256: C 257: C 258: C 259: B 260: A

14. Combine and Reactive Programming

261: A 262: C 263: C 264: A 265: C 266: D 267: D 268: D 269: B 270: A

271: C 272: C 273: C 274: D 275: C 276: C 277: A 278: B 279: A 280: D

15. Practical iOS Scenarios

281: A 282: B 283: C 284: D 285: D 286: B 287: A 288: A 289: B 290: A

291: A 292: A 293: D 294: A 295: C 296: D 297: A 298: D 299: B 300: B

16. Builds, Signing and App Distribution

301: A 302: C 303: D 304: C 305: B 306: C 307: C 308: C 309: A 310: B

311: A 312: C 313: D 314: D 315: C 316: A 317: D 318: B 319: C 320: C